



920 Trenton Frankford Road

A data centre that heats its neighbours

Supporting documents filed with the City of Quinte West
Interim Control By-law 26-131, Council meeting of October 7, 2026



Prepared by FirstBlock for Mayor, Council, City staff and residents
October 2, 2026

FirstBlock Inc. | firstblock.com | bd@firstblock.com
Vincent Paquet, co-founder, 819-247-2840 | Simon Rodrigue, 438-399-9563

Contents

#	Document	Page
1	The use is permitted: zoning confirmation from the City's Director of Planning (June 12, 2026)	3
2	The power is secured: Hydro One capacity confirmation, 18 MW at the existing connection (March 3, 2026)	4
3	Where the heat goes: the district heating network for greenhouses, farms and neighbouring buildings (FirstBlock and Ecosystem, March 2026)	6
4	Noise in one page: quieter than a refrigerator, compared with everyday sounds	8
5	How quiet is 45 dBA? Sound levels in context, with Ontario NPC-300 limits (TrueNorth Sustainable Infrastructure)	9
6	Noise Impact Study by HGC Engineering, P.Eng.: 34 to 41 dBA at the nearest homes, within Ontario limits (September 2026)	10
7	Above the flood line: surveyor's letter of conformity, IBW Surveyors (June 10, 2026)	31
8	The regulatory floodplain map in dispute: KGS Group / Lower Trent Conservation, sheet 112 (Feb. 2024), with FirstBlock's note	32
9	The ground as measured: topographic survey of the property by IBW Surveyors (May 29, 2026)	33
10	Flood level versus ground level: FirstBlock's comparison over the survey, no flood zone on the property	34
11	The containment wall already in place: original civil drawings, Glen Miller Power Inc. (sheets C01, C02, C03, C05)	35
12	Site civil engineering: grading, stormwater and servicing drawings, FDC (2026-019, signed)	39
13	Letter of support from Mayor Jim Harrison, City of Quinte West (December 4, 2025)	48
14	Letter of support from Joël Lightbound, Member of Parliament	49
15	What we are asking Council: proposed exemption to By-law 26-131 for projects already zoned and deemed complete	53

54 pages in total. Documents 3, 4, 8, 10 and 15 were prepared or marked up by FirstBlock, and document 5 by TrueNorth Sustainable Infrastructure; the underlying studies, map and survey are reproduced as issued by their authors. Documents 9, 11 and 12 are large-format drawings reduced to fit the page.

Date: Fri, Jun 12, 2026 at 2:31 PM

Subject: Fw: 869 / 920 Trenton -Frankford Road, City of Quinte West

To: Simon Rodrigue <simon@firstblock.com>, Pascal Perreault <pascal@firstblock.com>, Vincent Paquet <vincent@firstblock.com>

Hi,

Please see below.

Thank you,

Alsaa

All Season Fencing Ltd

From: Brian Jardine <brianj@quintewest.ca>

Sent: Friday, June 12, 2026 12:28 PM

Subject: 869 / 920 Trenton -Frankford Road, City of Quinte West

Good afternoon,

I understand from Ms. King of the Mayor / CAO's office that you are seeking zoning clarification on the above-mentioned property for the purposes of a meeting later today; and in advance of the arranged meeting with the City scheduled for June 19th. As you are aware, the subject properties are Zoned GM - General Industrial in the City's comprehensive Zoning By-law. We understand the proponent wishes to house a Data Centre within existing buildings on the property. While not explicitly listed within the range of uses within the GM zone, the City confirms that the proposed use is covered by the existing zone definitions, and is therefore permitted as a use within the GM zone.

Implementation of the use on the property is subject to the technical requirements already under discussion, and confirmed through a Site Plan Control process.

I trust this provides the clarification that you require at this point, and I look forward to our scheduled meeting next week.

Regards,

Brian

Brian Jardine

Director of Planning and Development Services

City of Quinte West

7 Creswell Drive, Trenton Ontario, K8V 5R6

E: brianj@quintewest.ca

T: 613-392-2841

Your working hours may be different from mine. Please do not feel that you have to respond outside of your normal working day.

Mar 3, 2026

2874059 Ontario Inc (All Season Fencing)
5 Bernard Long Rd
Trenton, On.
K8V 5P6

Attention: Moshe Yitzhack

Dear Moshe:

This letter confirms that Hydro One Networks Inc. ("**Hydro One**") has granted your facility located at 5 Bernard Long Road, Trenton, Ontario (the "**Facility**") a capacity increase, based on the information that you provided in the New Customer Connection information ("**NCCI**") form submitted on Aug 25, 2025 and the following information in particular:

Connection Information

Supply Station & Feeder: Sydney TS-M1
Existing Service (kW): 9900
Approved Required Service (kW): 18000
Motor Information: 150hp combined inrush of 1500kVA
Customer Account Number: 200288143622

Hydro One's approval of your requested capacity increase is subject to the following Connection Limitations and Requirements:

- (a) The facility has an assigned total capacity of **18000kW** at **0.85 PF**, equal to the highest rolling three-month average peak load in your approved NCCI form.
- (b) The highest rolling three-month average peak load of your facility shall not exceed the assigned capacity at any time without Hydro One's prior written approval.
- (c) Hydro One has performed the Connection Assessment based on the loading profile provided in your NCCI form, including consideration of the seasonality of your projected load. Significant load increases above the NCCI forecast for any time of year must be approved by Hydro One. In all cases, Hydro One reserves the right to limit the load taken by your facility to the monthly load forecast that was approved by Hydro One.
- (d) You have until January 2027 (the "**Assigned Capacity Date**") to use your assigned capacity. After the Assigned Capacity Date, the capacity assigned to your facility will be equal to your highest rolling three-month average peak load under normal operating conditions in the most recent three year period, and any unused assigned capacity will be cancelled and made available to other customers. Hydro One may, upon request and in its sole discretion, provide an extension, in writing, to the Assigned Capacity Date where circumstances warrant.
- (e) Starting of **150 hp** motor(s) with combined inrush of **1500kVA** does not exceed the flicker limit for Hydro One's distribution system, and is acceptable*.

*As stated in the Hydro One Conditions of Service section 2.3.3 [Customer Responsibilities], "If it is determined that unacceptable conditions are being caused by any Customer Equipment, the Customer shall take appropriate remedial action to correct the condition."

For reference, Hydro One's Conditions of Service can be found at the following link [Hydro One Conditions of Service](#).

If you have any questions about the above, please contact me at 613-585-2370.

Sincerely,

Sean Richards
Hydro One Networks Inc
Account Executive
Key Account Management
Cell: (613) 585-2370
Email: sean.richards@hydroone.com



Integrated Data Center and District Heating Network

FirstBlock and Ecosystem

Potential Project in Trenton, ON

March 23rd 2026



FirstBlock 

 **ecosystem**

Project Vision

FirstBlock plans to develop a data center at the former Sonoco paper mill in Trenton, Ontario, and to recover its waste heat to serve nearby buildings and a new greenhouse development. The project would include at a minimum thermal loop with flowing hot water to capture this heat and distribute it locally. This approach would directly reduce natural gas consumption, which would otherwise be used to heat these buildings, and improve the area's overall energy performance. The project would be developed in two phases, starting with a 16 MW data center in Phase 1, followed by a 17 MW expansion in Phase 2, for a total facility capacity of 33 MW.

Benefits

- Reuse of the data center's waste heat for heating needs
- Electricity used twice instead of once (data processing and heating)
- Data center sized/optimized to match heat demand
- Potential to recover and use up to 100% of rejected heat during peak periods
- Reduction of GHG emissions by 4,220 tonnes of CO₂ per year following phase 1 and 8,725 tonnes of CO₂ per year after phase 2, assuming the required heat would otherwise be supplied by natural gas.
- Supply of affordable, reliable, low-carbon heat for users
- Contribution to the development of an agri-food economic hub in Trenton

Proposed District Heating Network

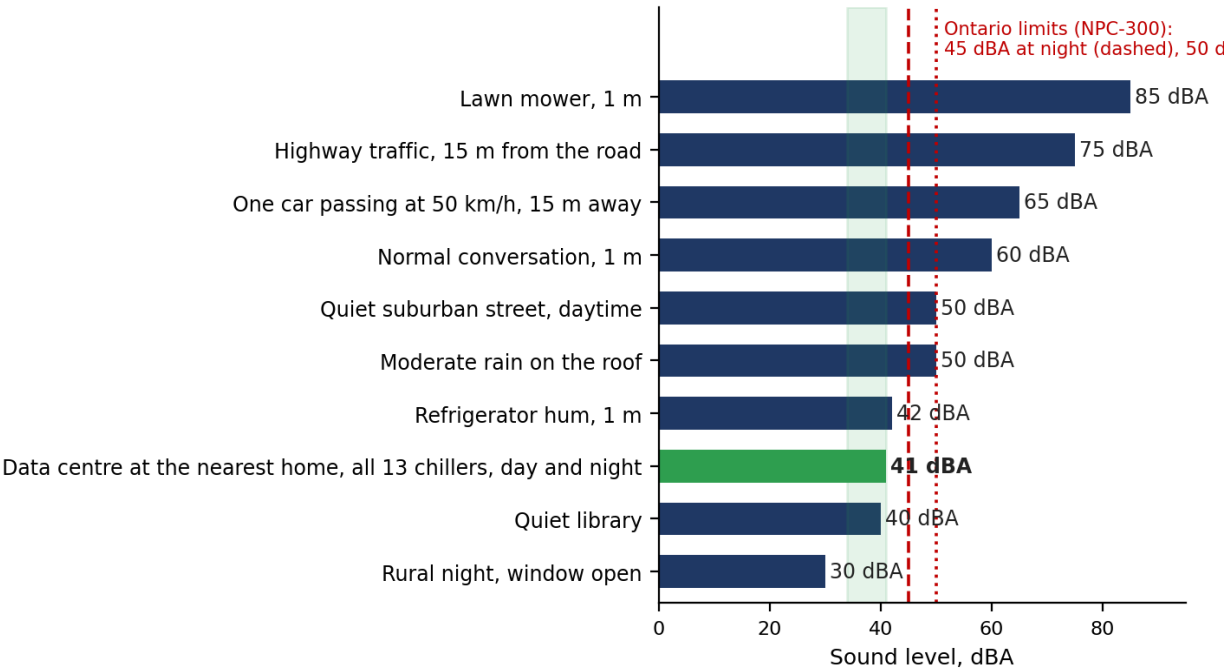


920 TRENTON FRANKFORD ROAD, QUINTE WEST

How loud will the data centre be? Quieter than a refrigerator.

An independent acoustic study by HGC Engineering (P.Eng., Ontario, September 9, 2026), filed with the City, predicts **34 to 41 dBA** at the seven nearest homes with all thirteen chillers running at once, day and night. Ontario's limits for this area (MECP guideline NPC-300) are **50 dBA by day and 45 dBA at night**. Every home is within the limit, with 4 to 11 dB to spare.

Everyday sounds compared with the data centre



What the study measured	Result	Limit
Normal operation, 13 chillers, worst-case hour, nearest homes (R1 to R7)	34 to 41 dBA	50 day / 45 night
Emergency generator test: one unit, one hour, once a month, daytime only	27 to 49 dBA	55 (daytime)
Vibration: servers, pumps and fans only, no press or heavy machinery	No vibration source	n/a

Three things worth knowing about decibels

The scale is logarithmic. A 10 dB increase sounds about twice as loud; a 3 dB change is barely noticeable. The gap between the data centre (41 dBA) and a single car passing on Trenton Frankford Road (about 65 dBA) is 24 dB: the car is roughly five times louder. **Within the limit does not mean silent.** On a very quiet night, a steady 35 to 41 dBA can be audible at the closest homes, in the range of a fridge in the next room. **The study is conservative.** It assumes all thirteen chillers at full output at the same time; free cooling for most of the year means fans run slower.

Our commitment

Independent sound measurements at the nearest homes after commissioning, with results made public. If any measured level exceeds the Ontario limit, we fix it at our cost. We are prepared to register this in the site plan agreement.

Sources: HGC Noise Vibration Acoustics, Noise Impact Study, 920 Trenton Frankford Road, September 9, 2026 (Tables 1 and 2; Class 2 area, NPC-300 exclusion limits); MECP Publication NPC-300. Everyday sound levels are typical published values (U.S. CDC/NIOSH, U.S. EPA, Health Canada) and vary with distance and conditions. Prepared by FirstBlock Inc., October 2026. Questions: bd@firstblock.com, 438-399-9563.

How Quiet Is 45 dBA?

TNSI Ontario data centres are designed to be **very quiet** at the nearest residential property line at night — similar to a kitchen fridge, but outside the home.

40 dBA



Quiet room, similar to a laptop fan under light usage

Ontario NPC-300 rural night limit

45 dBA



Between a quiet room and a quiet suburban evening, similar to a kitchen fridge

Ontario NPC-300 urban night limit

50 dBA



Outdoors in a quiet suburb

Ontario NPC-300 urban day limit

55 dBA



Conversation at 1 metre, similar to a typical home furnace

60 dBA



Normal conversation in a quiet room, similar to an outdoor air conditioner unit

65 dBA



Passenger car at 60 km/h, 20 m away, or an airplane near an airport



70 dBA



Inside a car on the highway, windows closed

80 dBA



Standing at the edge of a busy road

85 dBA



Diesel truck at 50 km/h, 20 m away

90 dBA



Gas lawn mower

100 dBA



ATV or motorcycle; earbuds at loud volume



TNSI Ontario data centres:



≤45 dBA

at the nearest residential property line at night

≤40 dBA in rural areas

Modelled by a third-party certified engineering firm.



Canadian communities are protected by rigorous standards.

Ontario's NPC-300 sets noise limits for industry, including data centres, to help keep communities quiet.



Every +10 dBA sounds about twice as loud.

Sound levels are measured on a logarithmic scale, so a 10 dBA increase is roughly perceived as twice as loud.



Our commitment

TNSI designs our Ontario data centres to operate well within applicable limits, with sound levels modelled by a third-party certified engineering firm.

September 9, 2026

First Block Inc.
Attn: Simon Rodrigue
5165 Rue John-Molson
Quebec, Quebec G1X 3X4

Via Email: simon@firstblock.com

RE: Noise Impact Study for Proposed Data Centre
920 Trenton Frankford Road, Quinte West, Ontario
HGC Project #: 02600694

HGC Noise Vibration Acoustics was retained by First Block Inc. to conduct a Noise Impact Study for a proposed data centre located at 920 Trenton Frankford Road in Quinte West, Ontario.

The assessment detailed herein is based on the site plan of the proposed data centre, sound emission levels provided by the equipment manufacturers, aerial imagery, information pertaining to operating hours, and a visit to the site and surrounding area by HGC in July 2026. The analysis includes an assessment of the noise impact anticipated at the nearest and most potentially impacted sound sensitive points of reception, conducted in accordance with the assessment methods and sound level limits contained in the Ontario Ministry of Environment, Conservation and Parks ("MECP") guideline, Publication NPC-300, as required by the City.

A computational acoustical model of the facility and the surrounding area was developed, in order to predict its sound levels at the nearby points of reception. The results indicate that the sound levels of the facility will be within the applicable limits of the City/MECP. The analysis and results are summarized in the following sections.

SITE DESCRIPTION AND NOISE SOURCES

A scaled area map is appended as Figure 1, which shows the location of the site and the sound sensitive receptors in the vicinity of the facility.

The data centre will consist of an existing industrial building which will be renovated. Typical operations at the facility can occur up to 24 hours per day, 7 days per week.

The subject lot and surrounding lands to the south, east, and west are zoned General Industrial. The lands to the north are zoned Light Rural. The nearest and most potentially impacted sound sensitive points of reception consist of existing homes on

lands zoned Residential Type 3 further to the west and south, and further to the east and north-east, across the Trent River.

Seven key assessment locations were chosen to represent the sound sensitive points of reception in the vicinity of the facility, shown as locations R1 through R7 in Figure 1, which are all existing single-family residences. Locations R1 and R5 represents the main floor windows of single-storey residences to the west and northeast of the site, respectively. Locations R2 to R4 and R6 represent the upper storey windows of two-storey residences. Locations R2 and R3 are located to the west of the site, R4 is located south of the site, and R6 is located to the east of the site. Location R7 represents the upper-storey window of an elevated single-storey residence to the east of the site.

The layout of the proposed data centre is illustrated on the site plan, which was prepared and provided by Fusion Consulting Ltd., and is included as Figure 2. The principal sources of sound will be thirteen chillers and six diesel-fired emergency generators, each paired with a transformer. Figure 3 shows the locations of each source.

ASSESSMENT CRITERIA

While the City of Quinte West maintains By-Law Number 08-127 "Noise Control By-law", the City relies on the noise assessment guidelines of the MECP for evaluating noise from "stationary" (fixed mechanical, commercial/industrial) sources of sound.

MECP Guideline NPC-300, "Environmental Noise Guideline Stationary and Transportation Sources – Approval and Planning" is the MECP guideline used for investigating Land Use Compatibility issues with regard to noise, and for establishing the applicable sound levels limits at surrounding sound sensitive points of reception. An industrial or commercial facility such as the proposed data centre is classified in MECP guidelines as a "stationary" source of sound for noise assessment purposes (versus sources such as traffic or construction, for example). In terms of the existing acoustical environment, the subject property is best categorized as a Class 2 (urban) Area under MECP guidelines, with background sound characteristically dominated by road traffic on Trenton Frankford Road and Glen Miller Road during daytime hours, with low evening and nighttime background sound levels defined the surrounding natural environment.

The MECP noise assessment guidelines draw a distinction between sound produced by traffic sources and that produced by stationary sources of sound. Sound level limits for stationary sources apply at noise sensitive points of reception (e.g., residences) and are set as the greater of either the applicable exclusion limit, or the minimum background sound level that occurs during the period corresponding to the operation of the source under assessment. Background sound is defined according to NPC-300 to include manmade noise such as road traffic or surrounding industry, but to exclude the sound of the facility under assessment.



NOISE



VIBRATION



ACOUSTICS

www.hgcacoustics.com

The exclusion limits in a Class 2 area are 50 dBA during daytime and evening hours (07:00 to 19:00 and 19:00 to 23:00, respectively) and 45 dBA during nighttime hours (23:00 to 07:00) at the plane of a window leading to a noise-sensitive space. For outdoor points of reception, the exclusionary minimum limits are 50 dBA during daytime and evening hours.

Sound emissions from emergency equipment operating in non-emergency situations, such as maintenance testing, are assessed independently of other stationary sources of noise and the applicable limits are 5 dB greater than the limits otherwise applicable to non-emergency sources. Since the emergency generators and corresponding transformers at the subject site are only tested during the daytime hours, the applicable daytime sound level limits are 55 dBA – i.e., 5 dBA greater than the otherwise applicable daytime limit of 50 dBA.

Road traffic data for Highway 401 at Glen Miller Road was obtained from the Ontario Ministry of Transportation in the form of quarter-hourly counts during November 2025 and can be found appended. The data was aggregated into hourly counts towards determining an hourly L_{EQ} at the assessment locations, and used in the MECP traffic modelling software *STAMSON*, version 5.04, which is commonly used throughout Ontario. The modelling results indicated that the hourly traffic sound levels from Highway 401 fall well below the exclusionary minimum limits during the quietest hours of the day, evening, and night, and observations conducted by HGC in the vicinity of the points of reception correlated well with the traffic noise modelling. Therefore, the exclusion limits for a Class 2 area of 50/45 dBA, as described above, have utilized for this study.

In addition, the MECP guidelines stipulate that the sound level impact during a “predicable worst-case hour” be considered. This is defined as an hour when a typically busy “planned and predictable mode of operation” occurs at the subject facility, coincident with a period of minimal background sound, which has been considered in the analysis presented, herein. Note that compliance with MECP criteria generally results in acceptable levels of sound at residential receptors although there may still be residual audibility during periods of low background sound.

Some types of sound have a special quality which may tend to increase their audibility and potential for disturbance or annoyance. For tonal sound, MECP guideline NPC-104, “*Sound Level Adjustments*”, stipulates that an adjustment of +5 dBA is to be added to the measured source level. A tonal sound is defined as one which has a “pronounced audible tonal quality such as a whine, screech, buzz or hum”. Tonal sources, to which the adjustment has been applied, are marked as such in Tables A1 in Appendix A.

PREDICTION METHODS AND RESULTS

Sound emission levels for the facility equipment were based on manufacturer’s sound data that was available for the chillers, while sound emission levels from similar units



NOISE



VIBRATION



ACOUSTICS

www.hgcacoustics.com

measured in the past by HGC were used for the generators and published estimation methods from reference acoustical engineering texts were used for the transformers based on the power capacity rating. All manufacturer's data can be found appended below. The assessment assumes that all non-emergency equipment, i.e., the thirteen chillers, operate simultaneously and continuously. Testing of the onsite emergency generators and corresponding transformers are scheduled to occur once per month for one hour per generator and transformer pair. Generators will be testing one at a time, with additional annual testing once per year with a duration of four hours. This assessment does not apply to the emergency generators and transformers operating in emergency situations, i.e., a power outage to the facility.

The sound emission levels (listed in Table A1 of Appendix A) and the site geometry shown in Figure 2 were used as input to a predictive computer model (Cadna/A version 2025), in order to determine the offsite sound levels at the receptors. Cadna/A is a software implementation of ISO Standard 9613-2 "Acoustics – Attenuation of Sound During Propagation Outdoors – Part 2: General Method of Calculation," which takes into account attenuation due to distance (geometrical spreading), shielding by intervening structures (such as buildings, where applicable), air attenuation and ground absorption. The operating times outlined above were assumed in determining the one-hour energy-equivalent sound level, L_{EQ} , for a predictable worst-case hour of operation.

The predicted sound levels, during predictable worst-case, typical operating conditions, are shown in Table 1, below, all of which comply with the applicable limits of the MECP. These results are also shown graphically as sound level contours, in Figure 4, below.

Table 1: Sound Level Predictions of Typical Operations, L_{EQ} [dBA]

Source	Point of Reception						
	R1	R2	R3	R4	R5	R6	R7
Sum of Thirteen Chillers	34	34	35	37	40	41	40
Total of All Sources	34	34	35	37	40	41	40
MECP Assessment Criteria (Day-Evening / Night)	50 / 45	50 / 45	50 / 45	50 / 45	50 / 45	50 / 45	50 / 45
Within Criteria?	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Of the six emergency generators and corresponding transformers, the offsite sound levels at the assessment locations are the greatest during testing of the west-most generator and transformer, EM-01 and EM-02. Thus, testing of EM-01 and EM-02 have been considered as the worst-case daytime testing of emergency equipment, with the predicted sound levels shown in Table 2, below, all of which comply with the applicable limits of the MECP. These results are also shown graphically as sound level contours, in Figure 5, below.

Table 2: Sound Level Predictions of Daytime Testing of Emergency Equipment, L_{EQ} [dBA]

Source	Point of Reception						
	R1	R2	R3	R4	R5	R6	R7
West-most Emergency Generator	42	48	49	27	37	36	31
West-most Transformer	18	25	16	--	18	18	8
Total of All Sources	42	48	49	27	37	34	31
MECP Assessment Criteria	55	55	55	55	55	55	55
Within Criteria?	Yes	Yes	Yes	Yes	Yes	Yes	Yes

CONCLUSIONS & RECOMMENDATIONS

The predicted sound levels are within the MECP assessment criteria, given the proposed site layout and equipment sound levels considered herein.

We trust the above is sufficient for your current purposes. If we can be of further assistance, please call.

Best regards,
HGC Noise Vibration Acoustics

Prepared by:


Nik Bennett, BEng

Reviewed by:


Robert D. Stevens, MASC, P.Eng.
Principal



Limitations

This document was prepared solely for the addressed party and titled project or named part thereof and should not be relied upon or used for any other project without obtaining prior written authorization from HGC Noise Vibration Acoustics (HGC). Further, the input of content from any document produced by HGC or related HGC intellectual property into any Artificial Intelligence tool is expressly prohibited. HGC accepts no responsibility or liability for any consequence of this document being used for a purpose other than for which it was commissioned. Any person or party using or relying on the document for such other purpose agrees and will by such use or reliance be taken to confirm their agreement to indemnify HGC for all loss or damage resulting therefrom. HGC accepts no responsibility or liability for this document to any person or party other than the party by whom it was commissioned.

Any conclusions and/or recommendations herein reflect the judgment of HGC based on information available at the time of preparation and were developed in good faith on information provided by others, as noted in the report, which has been assumed to be factual and accurate. Changed conditions or information occurring or becoming known after the date of this report could affect the results and conclusions presented.

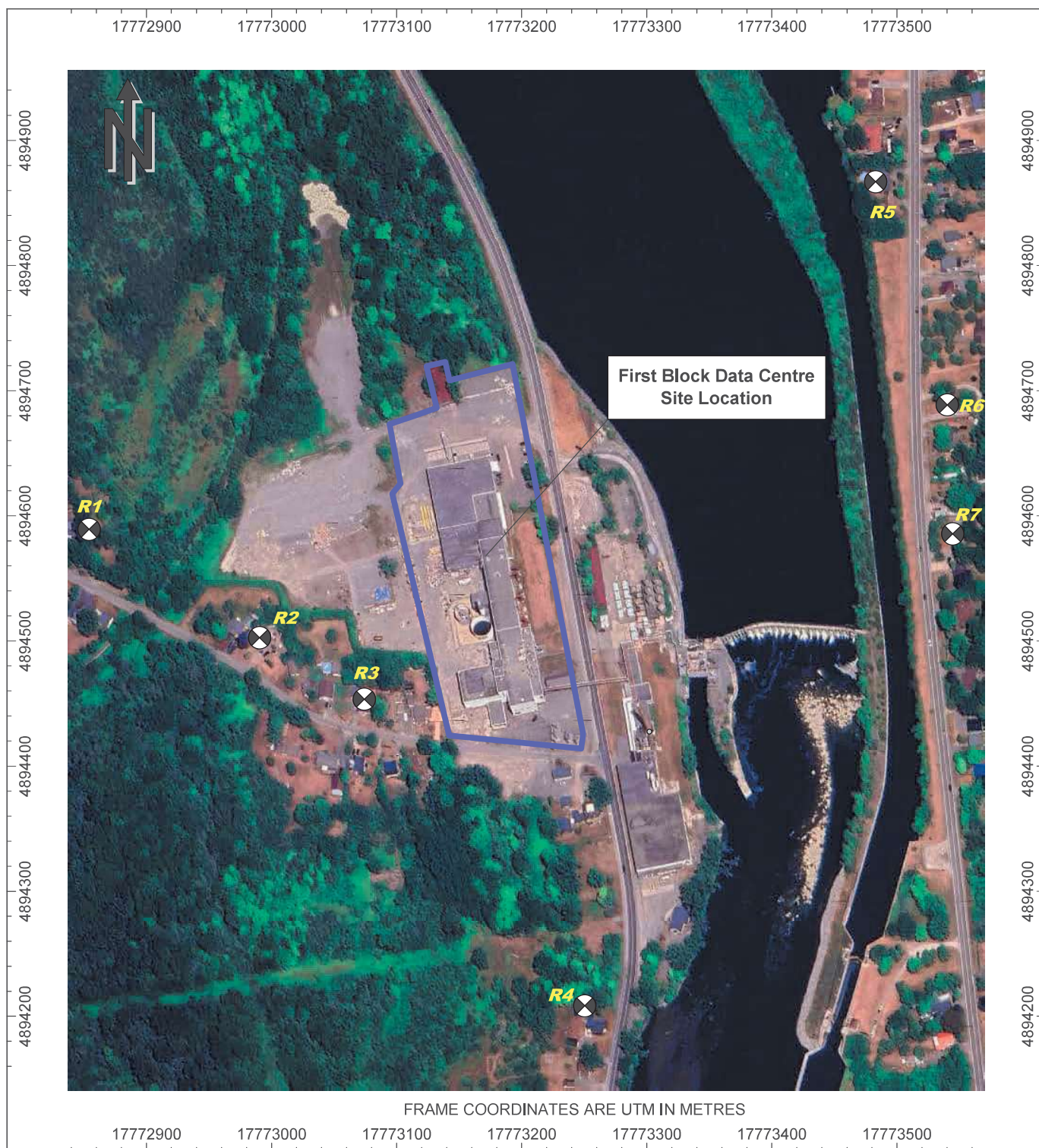


Figure 1: First Block Data Centre Site Location and Nearby Points of Reception

Figure 2: Proposed Data Centre Site Plan





Figure 3: Locations of Sound Sources

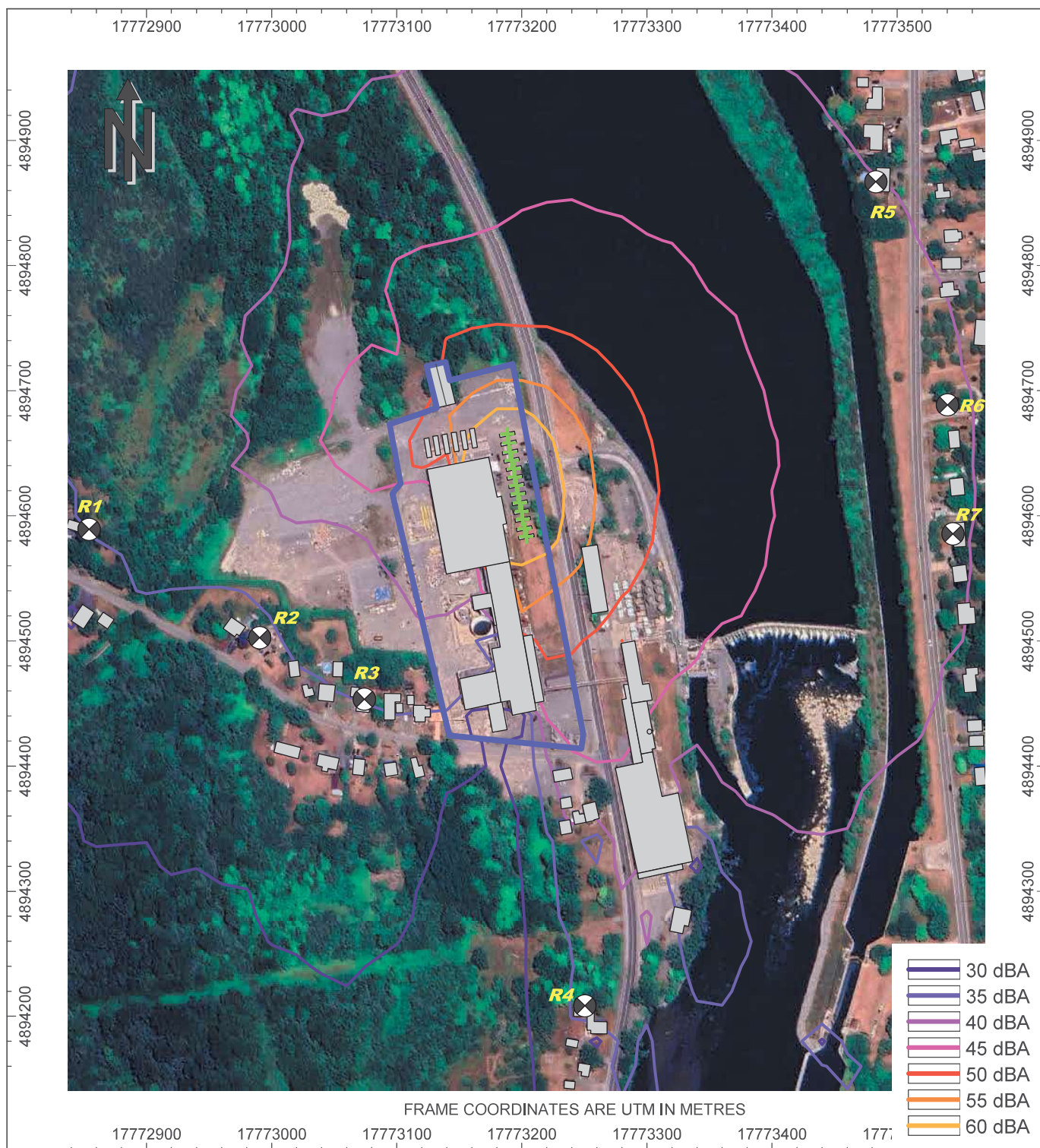


Figure 4: Sound Level Contours of Typical Operations, Leq [dBA]
Prediction Height = 4.5 m

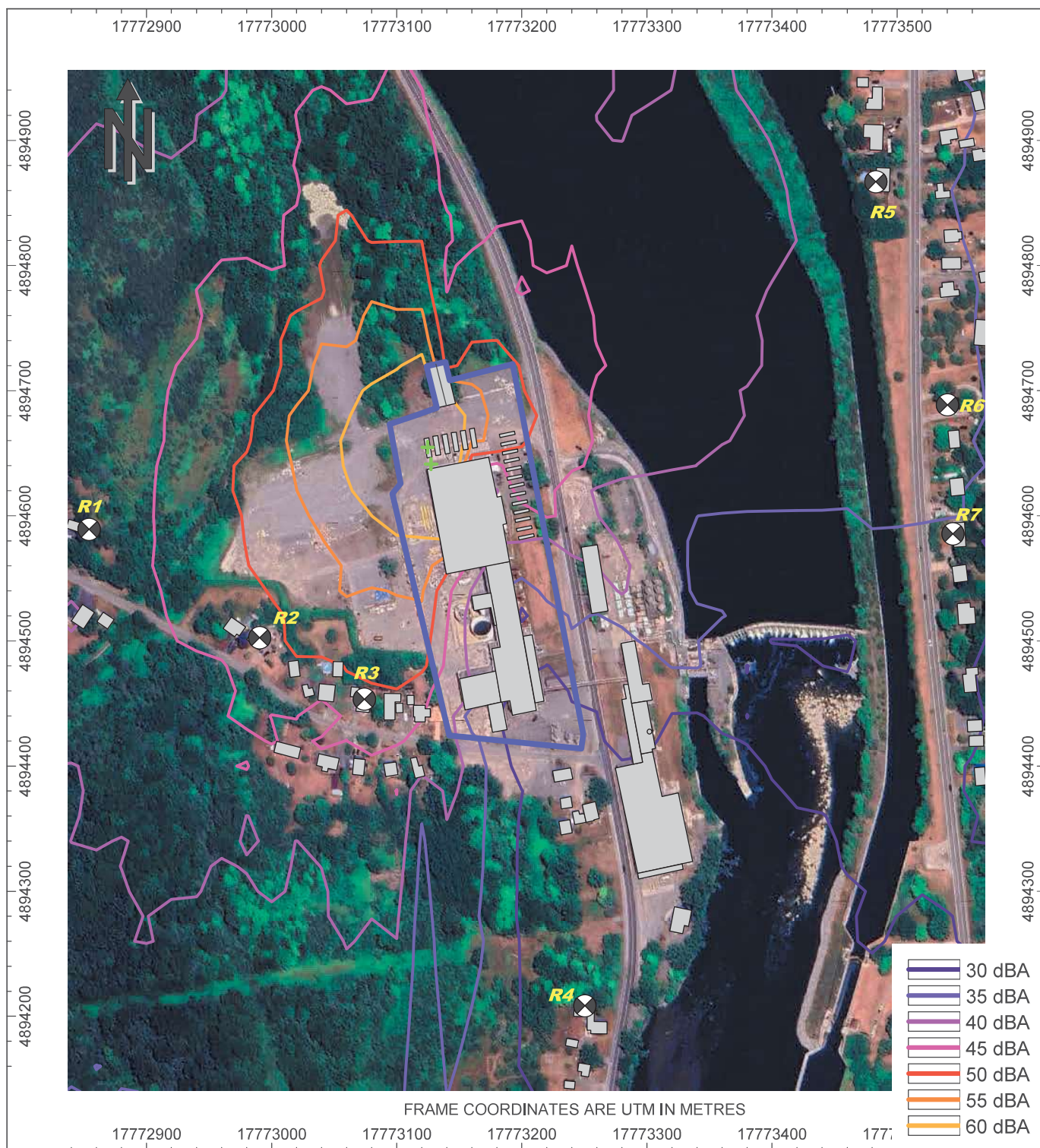


Figure 5: Sound Level contours of Worst-Case Emergency Equipment Testing
Daytime (07:00 - 19:00), Leq [dBA]
Prediction Height = 4.5 m

Appendix A

Acoustic Summary Tables



NOISE



VIBRATION



ACOUSTICS

www.hgcacoustics.com

Table A1: Noise Source Summary Table

Source ID	Source Description	Sound Power Level [dBA re 10 ⁻¹² W]	Source Location	Sound Characteristic	Noise Control Measure
NS-01	Chiller 1	94	O	S, T	U
NS-02	Chiller 2	94	O	S, T	U
NS-03	Chiller 3	94	O	S, T	U
NS-04	Chiller 4	94	O	S, T	U
NS-05	Chiller 5	94	O	S, T	U
NS-06	Chiller 6	94	O	S, T	U
NS-07	Chiller 7	94	O	S, T	U
NS-08	Chiller 8	94	O	S, T	U
NS-09	Chiller 9	94	O	S, T	U
NS-10	Chiller 10	94	O	S, T	U
NS-11	Chiller 11	94	O	S, T	U
NS-12	Chiller 12	94	O	S, T	U
NS-13	Chiller 13	94	O	S, T	U
EM-01	Generator 1	107	O	S	E, S
EM-02	Transformer 1	83	O	S, T	U
EM-03	Generator 2	107	O	S	E, S
EM-04	Transformer 2	83	O	S, T	U
EM-05	Generator 3	107	O	S	E, S
EM-06	Transformer 3	83	O	S, T	U
EM-07	Generator 4	107	O	S	E, S
EM-08	Transformer 4	83	O	S, T	U
EM-09	Generator 5	107	O	S	E, S
EM-10	Transformer 5	83	O	S, T	U
EM-11	Generator 6	107	O	S	E, S
EM-12	Transformer 6	83	O	S, T	U

Legend**Sound Characteristics**

S: Steady
 Q: Quasi-steady impulsive
 I: Impulsive
 B: Buzzing
 T: Tonal (+5 dBA penalty applied)
 C: Cyclically varying
 O: Occasional

Noise Control Measures

S: Silencer, Acoustic Louvre, Muffler
 A: Acoustic Lining, Plenum
 B: Barrier, Berm, Screening
 L: Lagging (Acoustical Wrapping)
 E: Acoustic Enclosure
 O: Other
 U: Currently Uncontrolled

Source Location

O: Outdoors
 I: Indoors

* Time weighted source. Reported sound power level does not include time weighted factor.

Table A2: Point of Reception Noise Impact Table

Source ID	Source Name	R1			R2			R3			R4			R5			R6			R7		
		Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	Dist [m]	LEQ [dBA]	LEQ [dBA]
NS-01	Chiller 1	340	27	254	26	239	24	460	24	359	28	355	29	367	27							
NS-02	Chiller 2	343	27	253	23	235	24	453	24	359	28	351	28	361	27							
NS-03	Chiller 3	342	27	249	22	230	24	447	24	363	29	352	30	360	29							
NS-04	Chiller 4	342	21	247	22	224	24	439	25	366	28	351	30	357	29							
NS-05	Chiller 5	343	20	245	22	220	24	433	25	368	29	350	30	354	29							
NS-06	Chiller 6	343	19	241	22	215	23	427	25	372	29	351	30	353	29							
NS-07	Chiller 7	346	19	241	22	211	23	419	25	374	29	348	30	348	28							
NS-08	Chiller 8	345	19	238	21	206	24	412	25	378	29	349	30	348	30							
NS-09	Chiller 9	345	18	234	22	200	24	405	24	383	29	351	30	347	30							
NS-10	Chiller 10	346	19	233	22	196	26	398	22	386	29	351	30	345	30							
NS-11	Chiller 11	346	21	230	25	191	28	391	26	391	29	352	30	344	30							
NS-12	Chiller 12	348	21	229	25	187	23	385	29	394	28	352	29	342	30							
NS-13	Chiller 13	348	22	227	25	182	21	378	29	398	28	354	29	342	30							
EM-01	Generator 1	276	42	201	48	207	49	464	27	418	37	419	36	428	31							
EM-02	Transformer 1	279	18	196	25	196	16	450	--	420	18	415	18	420	8							
EM-03	Generator 2	288	38	210	44	211	42	462	26	407	39	407	39	416	36							
EM-04	Transformer 2	285	18	202	22	199	10	450	--	414	20	409	17	414	9							
EM-05	Generator 3	292	37	214	43	214	35	463	26	403	38	403	39	412	38							
EM-06	Transformer 3	293	17	208	17	203	8	449	--	406	20	401	17	406	10							
EM-07	Generator 4	300	37	221	37	217	34	462	26	395	39	395	39	405	38							
EM-08	Transformer 4	302	17	215	11	207	7	449	--	399	20	392	17	399	9							
EM-09	Generator 5	308	37	228	37	222	34	462	26	388	39	387	39	397	38							
EM-10	Transformer 5	306	17	219	10	210	6	450	--	394	22	388	20	395	11							
EM-11	Generator 6	316	34	234	34	225	37	460	25	381	43	379	43	389	40							
EM-12	Transformer 6	318	17	229	9	217	6	449	--	384	19	376	20	383	16							

Note: Reported sound levels include all adjustment factors (time weighting, tonal penalty), as applicable.

Table A3a: Acoustic Assessment Summary Table - Typical Operations

Point of Reception	Point of Reception Description	Sound Level at Point of Reception, LEQ [dBA]			Performance Limit, LEQ [dBA]			Compliance with Performance Limit	Acoustical Classification Area	Verified by Acoustic Audit
		Day	Eve	Night	Day	Eve	Night			
R1	Single-Storey Residential Dwelling - 88 Bernard Long Rd	34	34	34	50	50	45	Yes/Yes/Yes	Class 2	No
R2	Two-Storey Residential Dwelling - 60 Bernard Long Rd	34	34	34	50	50	45	Yes/Yes/Yes	Class 2	No
R3	Two-Storey Residential Dwelling - 40 Bernard Long Rd	35	35	35	50	50	45	Yes/Yes/Yes	Class 2	No
R4	Two-Storey Residential Dwelling - 824 Trenton Frankford Rd	37	37	37	50	50	45	Yes/Yes/Yes	Class 2	No
R5	Single-Storey Residential Dwelling - 685 Glen Miller Rd	40	40	40	50	50	45	Yes/Yes/Yes	Class 2	No
R6	Two-Storey Residential Dwelling - 646 Glen Miller Rd	41	41	41	50	50	45	Yes/Yes/Yes	Class 2	No
R7	Elevated Single-Storey Residential Dwelling - 624 Glen Miller Rd	40	40	40	50	50	45	Yes/Yes/Yes	Class 2	No

Table A3b: Acoustic Assessment Summary Table - Worst-Case Emergency Equipment Testing

Point of Reception	Point of Reception Description	Sound Level at Point of Reception, LEQ [dBA]			Performance Limit, LEQ [dBA]			Compliance with Performance Limit	Acoustical Classification Area	Verified by Acoustic Audit
		Day	Eve	Night	Day	Eve	Night			
R1	Single-Storey Residential Dwelling - 88 Bernard Long Rd	42	--	--	55	--	--	Yes/--/--	Class 2	No
R2	Two-Storey Residential Dwelling - 60 Bernard Long Rd	48	--	--	55	--	--	Yes/--/--	Class 2	No
R3	Two-Storey Residential Dwelling - 40 Bernard Long Rd	49	--	--	55	--	--	Yes/--/--	Class 2	No
R4	Two-Storey Residential Dwelling - 824 Trenton Frankford Rd	27	--	--	55	--	--	Yes/--/--	Class 2	No
R5	Single-Storey Residential Dwelling - 685 Glen Miller Rd	37	--	--	55	--	--	Yes/--/--	Class 2	No
R6	Two-Storey Residential Dwelling - 646 Glen Miller Rd	36	--	--	55	--	--	Yes/--/--	Class 2	No
R7	Elevated Single-Storey Residential Dwelling - 624 Glen Miller Rd	31	--	--	55	--	--	Yes/--/--	Class 2	No

Appendix B

Summary of Calculations



NOISE



VIBRATION



ACOUSTICS

www.hgcacoustics.com

R1 Single-Storey Residential Dwelling - 88 Bernard Long Rd		17772854	4894569	94.3												
Src ID	Src Name	X	Y	Z	Lx	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	Cmet	RefI	Lr
NS-01	Chiller 1	17773186	4894665	94.3	94	61.7	0	0.0	1.3	2.7	1.2	0.0	0.0	0.0	0.0	27
NS-02	Chiller 2	17773190	4894658	94.3	94	61.7	0	0.0	1.3	2.9	1.4	0.0	0.0	0.0	0.0	27
NS-03	Chiller 3	17773190	4894652	94.3	94	61.7	0	0.0	1.3	2.9	1.4	0.0	0.0	0.0	0.0	27
NS-04	Chiller 4	17773192	4894644	94.2	94	61.7	0	0.0	1.6	9.4	0.5	0.0	0.0	0.0	0.0	21
NS-05	Chiller 5	17773194	4894638	94.3	94	61.7	0	0.0	1.8	10.7	0.4	0.0	0.0	0.0	0.0	20
NS-06	Chiller 6	17773194	4894632	94.4	94	61.7	0	0.0	1.8	11.2	0.4	0.0	0.0	0.0	0.0	19
NS-07	Chiller 7	17773198	4894624	94.5	94	61.7	0	0.0	1.9	11.1	0.4	0.0	0.0	0.0	0.0	19
NS-08	Chiller 8	17773198	4894618	94.7	94	61.7	0	0.0	1.9	10.9	0.4	0.0	0.0	0.0	0.0	19
NS-09	Chiller 9	17773198	4894611	94.8	94	61.8	0	0.0	1.9	11.9	0.4	0.0	0.0	0.0	0.0	18
NS-10	Chiller 10	17773200	4894604	95.0	94	61.8	0	0.0	1.9	11.5	0.4	0.0	0.0	0.0	0.0	19
NS-11	Chiller 11	17773200	4894597	96.0	94	61.8	0	0.0	1.9	9.4	0.5	0.0	0.0	0.0	0.0	21
NS-12	Chiller 12	17773202	4894590	96.0	94	61.8	0	0.0	2.0	8.5	0.5	0.0	0.0	0.0	0.0	21
NS-13	Chiller 13	17773202	4894584	96.0	94	61.9	0	0.0	2.0	8.4	0.5	0.0	0.0	0.0	0.0	22
EM-01	Generator 1	17773122	4894655	93.6	107	59.9	0	0.0	2.5	1.9	1.9	0.0	0.0	0.0	0.8	42
EM-02	Transformer 1	17773128	4894642	93.5	83	59.9	0	0.0	1.8	2.8	1.0	0.0	0.0	0.0	0.0	18
EM-03	Generator 2	17773134	4894656	93.6	107	60.1	0	0.0	3.6	6.3	0.4	0.0	0.0	0.0	1.8	38
EM-04	Transformer 2	17773134	4894644	93.5	83	60.1	0	0.0	1.8	2.8	1.0	0.0	0.0	0.0	0.0	18
EM-05	Generator 3	17773138	4894658	93.6	107	60.3	0	0.0	3.2	7.3	0.4	0.0	0.0	0.0	1.8	37
EM-06	Transformer 3	17773142	4894645	93.5	83	60.3	0	0.0	1.7	2.9	1.1	0.0	0.0	0.0	0.0	17
EM-07	Generator 4	17773146	4894659	93.6	107	60.6	0	0.0	3.0	8.1	0.5	0.0	0.0	0.0	2.0	37
EM-08	Transformer 4	17773150	4894647	93.5	83	60.6	0	0.0	2.9	1.6	1.1	0.0	0.0	0.0	0.0	17
EM-09	Generator 5	17773154	4894661	93.6	107	60.8	0	0.0	2.9	8.3	0.5	0.0	0.0	0.0	2.1	37
EM-10	Transformer 5	17773154	4894648	93.5	83	60.8	0	0.0	2.8	1.7	1.1	0.0	0.0	0.0	0.0	17
EM-11	Generator 6	17773162	4894661	93.6	107	61.0	0	0.0	2.8	8.4	0.5	0.0	0.0	0.0	0.0	34
EM-12	Transformer 6	17773166	4894650	93.5	83	61.0	0	0.0	2.7	1.7	1.1	0.0	0.0	0.0	0.0	17

R2 Two-Storey Residential Dwelling - 60 Bernard Long Rd		17772990	4894503	97.0												
Src ID	Src Name	X	Y	Z	LxD	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	CmetD	RefID	LrN
NS-01	Chiller 1	17773186	4894665	94.3	94	59.2	0	0.0	-0.2	10.0	0.4	0.0	0.0	0.0	1.1	26
NS-02	Chiller 2	17773190	4894658	94.3	94	59.1	0	0.0	-0.1	12.1	0.4	0.0	0.0	0.0	0.0	23
NS-03	Chiller 3	17773190	4894652	94.3	94	58.9	0	0.0	0.0	12.6	0.4	0.0	0.0	0.0	0.0	22
NS-04	Chiller 4	17773192	4894644	94.2	94	58.8	0	0.0	0.0	13.2	0.3	0.0	0.0	0.0	0.0	22
NS-05	Chiller 5	17773194	4894638	94.3	94	58.7	0	0.0	0.1	13.1	0.3	0.0	0.0	0.0	0.0	22
NS-06	Chiller 6	17773194	4894632	94.4	94	58.6	0	0.0	0.1	13.2	0.3	0.0	0.0	0.0	0.0	22
NS-07	Chiller 7	17773198	4894624	94.5	94	58.6	0	0.0	0.2	13.1	0.3	0.0	0.0	0.0	0.0	22
NS-08	Chiller 8	17773198	4894618	94.7	94	58.5	0	0.0	0.2	13.8	0.3	0.0	0.0	0.0	0.0	21
NS-09	Chiller 9	17773198	4894611	94.8	94	58.4	0	0.0	0.3	13.5	0.3	0.0	0.0	0.0	0.0	22
NS-10	Chiller 10	17773200	4894604	95.0	94	58.3	0	0.0	0.3	13.2	0.3	0.0	0.0	0.0	0.0	22
NS-11	Chiller 11	17773200	4894597	96.0	94	58.3	0	0.0	0.3	10.4	0.4	0.0	0.0	0.0	0.0	25
NS-12	Chiller 12	17773202	4894590	96.0	94	58.2	0	0.0	0.3	10.4	0.4	0.0	0.0	0.0	0.0	25
NS-13	Chiller 13	17773202	4894584	96.0	94	58.2	0	0.0	0.4	10.3	0.4	0.0	0.0	0.0	0.0	25
EM-01	Generator 1	17773122	4894655	93.6	107	57.1	0	0.0	-0.8	1.5	1.6	0.0	0.0	0.0	0.7	48
EM-02	Transformer 1	17773128	4894642	93.5	83	56.8	0	0.0	-0.9	1.8	0.9	0.0	0.0	0.0	0.0	25
EM-03	Generator 2	17773134	4894656	93.6	107	57.4	0	0.0	0.0	7.1	0.5	0.0	0.0	0.0	1.8	44
EM-04	Transformer 2	17773134	4894644	93.5	83	57.1	0	0.0	-0.4	10.4	0.4	0.0	0.0	0.0	6.2	22
EM-05	Generator 3	17773138	4894658	93.6	107	57.6	0	0.0	-0.1	6.9	0.6	0.0	0.0	0.0	0.7	43
EM-06	Transformer 3	17773142	4894645	93.5	83	57.3	0	0.0	-0.5	13.8	0.4	0.0	0.0	0.0	4.7	17
EM-07	Generator 4	17773146	4894659	93.6	107	57.9	0	0.0	0.0	13.7	0.4	0.0	0.0	0.0	2.1	37
EM-08	Transformer 4	17773150	4894647	93.5	83	57.6	0	0.0	-0.6	15.3	0.4	0.0	0.0	0.0	0.0	11
EM-09	Generator 5	17773154	4894661	93.6	107	58.1	0	0.0	-0.3	14.1	0.4	0.0	0.0	0.0	2.2	37
EM-10	Transformer 5	17773154	4894648	93.5	83	57.9	0	0.0	-0.7	16.0	0.4	0.0	0.0	0.0	0.0	10
EM-11	Generator 6	17773162	4894661	93.6	107	58.4	0	0.0	-0.4	14.3	0.4	0.0	0.0	0.0	0.0	34
EM-12	Transformer 6	17773166	4894650	93.5	83	58.1	0	0.0	-0.8	16.5	0.4	0.0	0.0	0.0	0.0	9

R3 Two-Storey Residential Dwelling - 40 Bernard Long Rd		17773074	4894454	97.5												
Src ID	Src Name	X	Y	Z	LxD	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	CmetD	RefID	LrN
NS-01	Chiller 1	17773186	4894665	94.3	94	58.6	0	0.0	-0.3	11.7	0.4	0.0	0.0	0.0	0.0	24
NS-02	Chiller 2	17773190	4894658	94.3	94	58.4	0	0.0	-0.2	11.9	0.4	0.0	0.0	0.0	0.0	24
NS-03	Chiller 3	17773190	4894652	94.3	94	58.2	0	0.0	-0.2	12.2	0.3	0.0	0.0	0.0	0.0	24
NS-04	Chiller 4	17773192	4894644	94.2	94	58.0	0	0.0	-0.1	12.5	0.3	0.0	0.0	0.0	0.0	24
NS-05	Chiller 5	17773194	4894638	94.3	94	57.8	0	0.0	-0.1	12.4	0.3	0.0	0.0	0.0	0.0	24
NS-06	Chiller 6	17773194	4894632	94.4	94	57.6	0	0.0	-0.1	13.6	0.3	0.0	0.0	0.0	0.0	23
NS-07	Chiller 7	17773198	4894624	94.5	94	57.4	0	0.0	-0.1	13.5	0.3	0.0	0.0	0.0	0.0	23
NS-08	Chiller 8	17773198	4894618	94.7	94	57.2	0	0.0	-0.1	13.2	0.3	0.0	0.0	0.0	0.0	24
NS-09	Chiller 9	17773198	4894611	94.8	94	57.0	0	0.0	-0.1	12.9	0.3	0.0	0.0	0.0	0.0	24
NS-10	Chiller 10	17773200	4894604	95.0	94	56.8	0	0.0	-0.1	11.7	0.3	0.0	0.0	0.0	0.0	26
NS-11	Chiller 11	17773200	4894597	96.0	94	56.6	0	0.0	-0.2	9.9	0.3	0.0	0.0	0.0	0.0	28
NS-12	Chiller 12	17773202	4894590	96.0	94	56.4	0	0.0	0.1	14.4	0.2	0.0	0.0	0.0	0.0	23
NS-13	Chiller 13	17773202	4894584	96.0	94	56.3	0	0.0	0.1	16.7	0.2	0.0	0.0	0.0	0.0	21
EM-01	Generator 1	17773122	4894655	93.6	107	57.3	0	0.0	-0.9	0.0	1.3	0.0	0.0	0.0	0.0	49
EM-02	Transformer 1	17773128	4894642	93.5	83	56.8	0	0.0	-0.8	10.8	0.4	0.0	0.0	0.0	0.0	16
EM-03	Generator 2	17773134	4894656	93.6	107	57.5	0	0.0	-0.4	7.4	0.5	0.0	0.0	0.0	0.0	42
EM-04	Transformer 2	17773134	4894644	93.5	83	57.0	0	0.0	-0.7	17.1	0.4	0.0	0.0	0.0	0.0	10
EM-05	Generator 3	17773138	4894658	93.6	107	57.6	0	0.0	-0.3	14.5	0.4	0.0	0.0	0.0	0.0	35
EM-06	Transformer 3	17773142	4894645	93.5	83	57.1	0	0.0	-0.7	18.8	0.4	0.0	0.0	0.0	0.0	8
EM-07	Generator 4	17773146	4894659	93.6	107	57.8	0	0.0	-0.4	15.0	0.4	0.0	0.0	0.0	0.0	34
EM-08	Transformer 4	17773150	4894647	93.5	83	57.3	0	0.0	-0.8	19.5	0.4	0.0	0.0	0.0	0.0	7
EM-09	Generator 5	17773154	4894661	93.6	107	57.9	0	0.0	-0.5	15.1	0.4	0.0	0.0	0.0	0.0	34
EM-10	Transformer 5	17773154	4894648	93.5	83	57.5	0	0.0	-0.8	19.8	0.4	0.0	0.0	0.0	0.0	6
EM-11	Generator 6	17773162	4894661	93.6	107	58.1	0	0.0	-0.5	15.5	0.4	0.0	0.0	0.0	3.2	37
EM-12	Transformer 6	17773166	4894650	93.5	83	57.7	0	0.0	-0.8	20.1	0.5	0.0	0.0	0.0	0.0	6

R4 Two-Storey Residential Dwelling -

Src ID	Src Name	X	Y	Z	Lx	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	Cmetd	Refl	Lr
NS-02	Chiller 2	17773190	4894658	94.3	94	64.1	0	0.0	0.2	4.4	1.2	0.0	0.0	0.0	0.0	24
NS-03	Chiller 3	17773190	4894652	94.3	94	64.0	0	0.0	0.2	4.6	1.1	0.0	0.0	0.0	0.0	24
NS-04	Chiller 4	17773192	4894644	94.2	94	63.9	0	0.0	0.2	4.6	1.2	0.0	0.0	0.0	0.0	25
NS-05	Chiller 5	17773194	4894638	94.3	94	63.7	0	0.0	0.1	4.2	1.3	0.0	0.0	0.0	0.0	25
NS-06	Chiller 6	17773194	4894632	94.4	94	63.6	0	0.0	0.1	4.4	1.2	0.0	0.0	0.0	0.0	25
NS-07	Chiller 7	17773198	4894624	94.5	94	63.5	0	0.0	0.2	4.7	1.1	0.0	0.0	0.0	0.0	25
NS-08	Chiller 8	17773198	4894618	94.7	94	63.3	0	0.0	0.3	5.0	1.1	0.0	0.0	0.0	0.0	25
NS-09	Chiller 9	17773198	4894611	94.8	94	63.2	0	0.0	0.5	5.6	0.9	0.0	0.0	0.0	0.0	24
NS-10	Chiller 10	17773200	4894604	95.0	94	63.0	0	0.0	0.8	7.2	0.7	0.0	0.0	0.0	0.0	22
NS-11	Chiller 11	17773200	4894597	96.0	94	62.9	0	0.0	0.1	3.5	1.4	0.0	0.0	0.0	0.0	26
NS-12	Chiller 12	17773202	4894590	96.0	94	62.7	0	0.0	0.0	1.0	1.3	0.0	0.0	0.0	0.0	29
NS-13	Chiller 13	17773202	4894584	96.0	94	62.5	0	0.0	0.1	1.1	1.2	0.0	0.0	0.0	0.0	29
EM-01	Generator 1	17773122	4894655	93.6	107	64.3	0	0.0	0.3	14.6	0.8	0.0	0.0	0.0	0.0	27
EM-02	Transformer 1	17773128	4894642	93.5	83	64.1	0	0.0	-0.5	20.2	0.9	0.0	0.0	0.0	0.0	--
EM-03	Generator 2	17773134	4894656	93.6	107	64.3	0	0.0	0.3	16.0	0.8	0.0	0.0	0.0	0.0	26
EM-04	Transformer 2	17773134	4894644	93.5	83	64.1	0	0.0	-0.5	21.1	1.0	0.0	0.0	0.0	0.0	--
EM-05	Generator 3	17773138	4894658	93.6	107	64.3	0	0.0	0.3	16.5	0.8	0.0	0.0	0.0	1.2	26
EM-06	Transformer 3	17773142	4894645	93.5	83	64.1	0	0.0	-0.5	21.2	1.0	0.0	0.0	0.0	0.0	--
EM-07	Generator 4	17773146	4894659	93.6	107	64.3	0	0.0	0.3	16.6	0.8	0.0	0.0	0.0	1.2	26
EM-08	Transformer 4	17773150	4894647	93.5	83	64.1	0	0.0	-0.5	21.2	1.0	0.0	0.0	0.0	0.0	--
EM-09	Generator 5	17773154	4894661	93.6	107	64.3	0	0.0	0.3	16.8	0.8	0.0	0.0	0.0	1.5	26
EM-10	Transformer 5	17773154	4894648	93.5	83	64.1	0	0.0	-0.5	21.1	1.0	0.0	0.0	0.0	0.0	--
EM-11	Generator 6	17773162	4894661	93.6	107	64.3	0	0.0	0.2	16.3	0.8	0.0	0.0	0.0	0.0	25
EM-12	Transformer 6	17773166	4894650	93.5	83	64.1	0	0.0	-0.5	20.6	1.0	0.0	0.0	0.0	0.0	--

R5 Single-Storey Residential Dwelling - 685 Glen Miller Rd		17773482	4894867	98.5												
Src ID	Src Name	X	Y	Z	LxD	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	Cmetd	ReflID	LrN
NS-01	Chiller 1	17773186	4894665	94.3	94	62.1	0	0.0	-1.5	4.1	1.5	0.0	0.0	0.0	0.0	28
NS-02	Chiller 2	17773190	4894658	94.3	94	62.1	0	0.0	-1.3	4.1	1.5	0.0	0.0	0.0	0.0	28
NS-03	Chiller 3	17773190	4894652	94.3	94	62.2	0	0.0	-1.2	4.0	1.5	0.0	0.0	0.0	1.4	29
NS-04	Chiller 4	17773192	4894644	94.2	94	62.3	0	0.0	-1.1	4.0	1.6	0.0	0.0	0.0	0.0	28
NS-05	Chiller 5	17773194	4894638	94.3	94	62.3	0	0.0	-1.0	3.9	1.6	0.0	0.0	0.0	1.9	29
NS-06	Chiller 6	17773194	4894632	94.4	94	62.4	0	0.0	-0.8	3.9	1.6	0.0	0.0	0.0	1.9	29
NS-07	Chiller 7	17773198	4894624	94.5	94	62.5	0	0.0	-0.8	3.8	1.6	0.0	0.0	0.0	2.0	29
NS-08	Chiller 8	17773198	4894618	94.7	94	62.6	0	0.0	-0.7	3.8	1.6	0.0	0.0	0.0	2.0	29
NS-09	Chiller 9	17773198	4894611	94.8	94	62.7	0	0.0	-0.7	3.8	1.6	0.0	0.0	0.0	2.2	29
NS-10	Chiller 10	17773200	4894604	95.0	94	62.8	0	0.0	-0.6	3.8	1.6	0.0	0.0	0.0	2.0	29
NS-11	Chiller 11	17773200	4894597	96.0	94	62.8	0	0.0	-0.6	3.8	1.6	0.0	0.0	0.0	2.0	29
NS-12	Chiller 12	17773202	4894590	96.0	94	62.9	0	0.0	-0.5	3.7	1.7	0.0	0.0	0.0	2.0	28
NS-13	Chiller 13	17773202	4894584	96.0	94	63.0	0	0.0	-0.5	3.7	1.7	0.0	0.0	0.0	1.8	28
EM-01	Generator 1	17773122	4894655	93.6	107	63.4	0	0.0	-1.1	7.6	1.1	0.0	0.0	0.0	1.2	37
EM-02	Transformer 1	17773128	4894642	93.5	83	63.5	0	0.0	-1.4	4.7	1.3	0.0	0.0	0.0	2.7	18
EM-03	Generator 2	17773134	4894656	93.6	107	63.2	0	0.0	-1.0	7.4	1.1	0.0	0.0	0.0	2.9	39
EM-04	Transformer 2	17773134	4894644	93.5	83	63.3	0	0.0	-1.4	4.6	1.2	0.0	0.0	0.0	4.6	20
EM-05	Generator 3	17773138	4894658	93.6	107	63.1	0	0.0	-0.8	7.5	1.0	0.0	0.0	0.0	2.4	38
EM-06	Transformer 3	17773142	4894645	93.5	83	63.2	0	0.0	-1.4	4.3	1.2	0.0	0.0	0.0	3.6	20
EM-07	Generator 4	17773146	4894659	93.6	107	62.9	0	0.0	-0.9	7.4	1.0	0.0	0.0	0.0	2.2	39
EM-08	Transformer 4	17773150	4894647	93.5	83	63.0	0	0.0	-1.3	3.2	1.1	0.0	0.0	0.0	3.0	20
EM-09	Generator 5	17773154	4894661	93.6	107	62.8	0	0.0	-1.2	7.4	1.0	0.0	0.0	0.0	2.1	39
EM-10	Transformer 5	17773154	4894648	93.5	83	62.9	0	0.0	-1.9	1.4	1.5	0.0	0.0	0.0	2.1	22
EM-11	Generator 6	17773162	4894661	93.6	107	62.6	0	0.0	-2.0	1.4	2.2	0.0	0.0	0.0	0.7	43
EM-12	Transformer 6	17773166	4894650	93.5	83	62.7	0	0.0	-1.2	2.8	1.3	0.0	0.0	0.0	1.4	19

R6 Two-Storey Residential Dwelling - 646 Glen Miller Rd		17773540	4894689	101.5												
Src ID	Src Name	X	Y	Z	LxD	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	Cmetd	ReflID	LrN
NS-01	Chiller 1	17773186	4894665	94.3	94	61.9	0	0.0	-0.7	3.7	1.5	0.0	0.0	0.0	1.3	29
NS-02	Chiller 2	17773190	4894658	94.3	94	61.9	0	0.0	-0.7	3.7	1.5	0.0	0.0	0.0	0.0	28
NS-03	Chiller 3	17773190	4894652	94.3	94	61.9	0	0.0	-0.6	3.7	1.5	0.0	0.0	0.0	1.9	30
NS-04	Chiller 4	17773192	4894644	94.2	94	61.9	0	0.0	-0.6	3.6	1.5	0.0	0.0	0.0	1.9	30
NS-05	Chiller 5	17773194	4894638	94.3	94	61.9	0	0.0	-0.5	3.6	1.5	0.0	0.0	0.0	1.9	30
NS-06	Chiller 6	17773194	4894632	94.4	94	61.9	0	0.0	-0.5	3.6	1.5	0.0	0.0	0.0	1.9	30
NS-07	Chiller 7	17773198	4894624	94.5	94	61.9	0	0.0	-0.4	3.6	1.4	0.0	0.0	0.0	1.9	30
NS-08	Chiller 8	17773198	4894618	94.7	94	61.9	0	0.0	-0.4	3.6	1.4	0.0	0.0	0.0	1.9	30
NS-09	Chiller 9	17773198	4894611	94.8	94	61.9	0	0.0	-0.3	3.5	1.4	0.0	0.0	0.0	1.9	30
NS-10	Chiller 10	17773200	4894604	95.0	94	61.9	0	0.0	-0.3	3.5	1.4	0.0	0.0	0.0	1.9	30
NS-11	Chiller 11	17773200	4894597	96.0	94	61.9	0	0.0	-0.3	3.5	1.4	0.0	0.0	0.0	1.9	30
NS-12	Chiller 12	17773202	4894590	96.0	94	61.9	0	0.0	-0.2	3.5	1.4	0.0	0.0	0.0	1.9	29
NS-13	Chiller 13	17773202	4894584	96.0	94	62.0	0	0.0	-0.2	3.5	1.4	0.0	0.0	0.0	1.9	29
EM-01	Generator 1	17773122	4894655	93.6	107	63.4	0	0.0	-1.7	8.3	1.0	0.0	0.0	0.0	0.0	36
EM-02	Transformer 1	17773128	4894642	93.5	83	63.4	0	0.0	-2.2	2.6	1.2	0.0	0.0	0.0	0.0	18
EM-03	Generator 2	17773134	4894656	93.6	107	63.3	0	0.0	-1.7	8.2	1.0	0.0	0.0	0.0	2.5	39
EM-04	Transformer 2	17773134	4894644	93.5	83	63.2	0	0.0	-2.3	4.6	1.3	0.0	0.0	0.0	0.0	17
EM-05	Generator 3	17773138	4894658	93.6	107	63.1	0	0.0	-1.6	8.3	1.0	0.0	0.0	0.0	2.6	39
EM-06	Transformer 3	17773142	4894645	93.5	83	63.1	0	0.0	-2.3	4.7	1.3	0.0	0.0	0.0	0.0	17
EM-07	Generator 4	17773146	4894659	93.6	107	62.9	0	0.0	-1.6	8.2	1.0	0.0	0.0	0.0	2.5	39
EM-08	Transformer 4	17773150	4894647	93.5	83	62.9	0	0.0	-2.2	4.7	1.3	0.0	0.0	0.0	0.0	17
EM-09	Generator 5	17773154	4894661	93.6	107	62.8	0	0.0	-1.5	8.3	1.0	0.0	0.0	0.0	2.6	39
EM-10	Transformer 5	17773154	4894648	93.5	83	62.7	0	0.0	-2.1	1.8	1.2	0.0	0.0	0.0	0.0	20
EM-11	Generator 6	17773162	4894661	93.6	107	62.6	0	0.0	-1.7	1.8	1.9	0.0	0.0	0.0	0.9	43
EM-12	Transformer 6	17773166	4894650	93.5	83	62.6	0	0.0	-2.0	1.8	1.2	0.0	0.0	0.0	0.0	20

R7 Elevated Single-Storey Residential Dwelling - 624 Glen Miller Rd					17773544	4894586	99.5												
SrcID	Src Name	X	Y	Z	LxD	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	CmetD	RefID	LrN			
NS-01	Chiller 1	17773186	4894665	94.3	94	62.2	0	0.0	0.0	3.3	1.5	0.0	0.0	0.0	0.0	27			
NS-02	Chiller 2	17773190	4894658	94.3	94	62.2	0	0.0	0.0	3.3	1.5	0.0	0.0	0.0	0.0	27			
NS-03	Chiller 3	17773190	4894652	94.3	94	62.1	0	0.0	0.0	3.3	1.5	0.0	0.0	0.0	1.9	29			
NS-04	Chiller 4	17773192	4894644	94.2	94	62.1	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	1.9	29			

Src ID	Src Name	X	Y	Z	Lx	Adiv	K0	Dc	Agnd	Abar	Aatm	Afol	Ahous	Cmet	Refl	Lr
NS-05	Chiller 5	17773194	4894638	94.3	94	62.0	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	1.9	29
NS-06	Chiller 6	17773194	4894632	94.4	94	62.0	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	1.9	29
NS-07	Chiller 7	17773198	4894624	94.5	94	61.9	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	0.6	28
NS-08	Chiller 8	17773198	4894618	94.7	94	61.9	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	1.9	30
NS-09	Chiller 9	17773198	4894611	94.8	94	61.8	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	1.9	30
NS-10	Chiller 10	17773200	4894604	95.0	94	61.8	0	0.0	0.1	3.2	1.4	0.0	0.0	0.0	1.9	30
NS-11	Chiller 11	17773200	4894597	96.0	94	61.7	0	0.0	0.2	3.2	1.4	0.0	0.0	0.0	1.9	30
NS-12	Chiller 12	17773202	4894590	96.0	94	61.7	0	0.0	0.2	3.2	1.4	0.0	0.0	0.0	1.9	30
NS-13	Chiller 13	17773202	4894584	96.0	94	61.7	0	0.0	0.2	3.2	1.4	0.0	0.0	0.0	1.9	30
EM-01	Generator 1	17773122	4894655	93.6	107	63.6	0	0.0	-0.4	12.2	0.8	0.0	0.0	0.0	0.0	31
EM-02	Transformer 1	17773128	4894642	93.5	83	63.5	0	0.0	-0.8	11.5	0.8	0.0	0.0	0.0	0.0	8
EM-03	Generator 2	17773134	4894656	93.6	107	63.4	0	0.0	-0.6	7.8	1.1	0.0	0.0	0.0	1.0	36
EM-04	Transformer 2	17773134	4894644	93.5	83	63.3	0	0.0	-0.8	11.1	0.8	0.0	0.0	0.0	0.0	9
EM-05	Generator 3	17773138	4894658	93.6	107	63.3	0	0.0	-0.5	8.0	1.1	0.0	0.0	0.0	2.8	38
EM-06	Transformer 3	17773142	4894645	93.5	83	63.2	0	0.0	-0.8	10.3	0.8	0.0	0.0	0.0	0.0	10
EM-07	Generator 4	17773146	4894659	93.6	107	63.1	0	0.0	-0.6	7.2	1.0	0.0	0.0	0.0	2.4	38
EM-08	Transformer 4	17773150	4894647	93.5	83	63.0	0	0.0	-0.8	10.9	0.8	0.0	0.0	0.0	0.0	9
EM-09	Generator 5	17773154	4894661	93.6	107	63.0	0	0.0	-0.5	7.6	1.0	0.0	0.0	0.0	2.5	38
EM-10	Transformer 5	17773154	4894648	93.5	83	62.9	0	0.0	-0.9	9.1	0.8	0.0	0.0	0.0	0.0	11
EM-11	Generator 6	17773162	4894661	93.6	107	62.8	0	0.0	-0.8	3.7	2.3	0.0	0.0	0.0	1.5	40
EM-12	Transformer 6	17773166	4894650	93.5	83	62.7	0	0.0	-1.0	4.0	1.3	0.0	0.0	0.0	0.0	16

Where: $L_r = L_x - A_{div} + K_0 + D_c - A_{gnd} - A_{bar} - A_{atm} - A_{fol} - A_{hous} + C_{met} + Refl$

Appendix C

Manufacturer's Data



NOISE

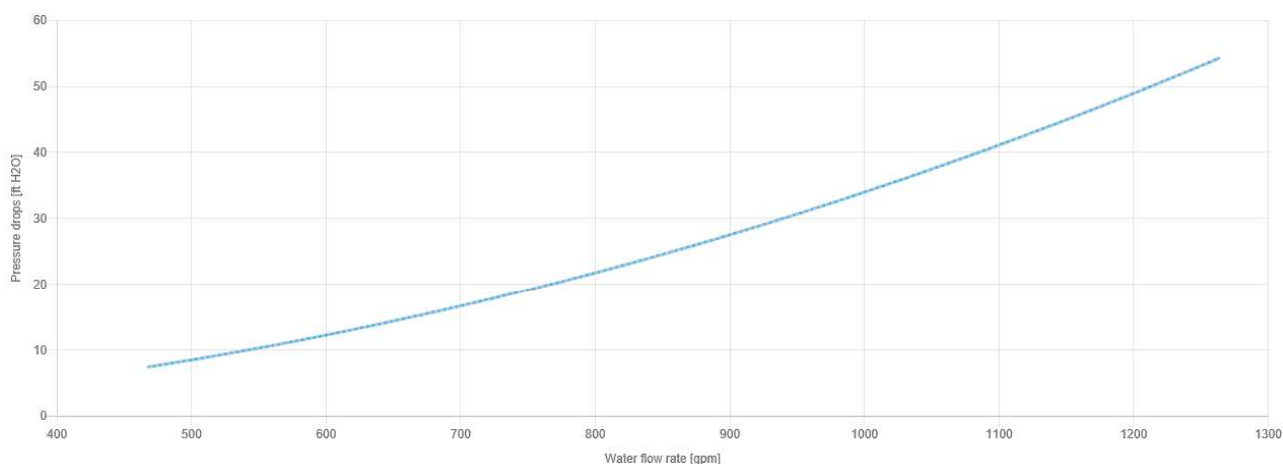


VIBRATION



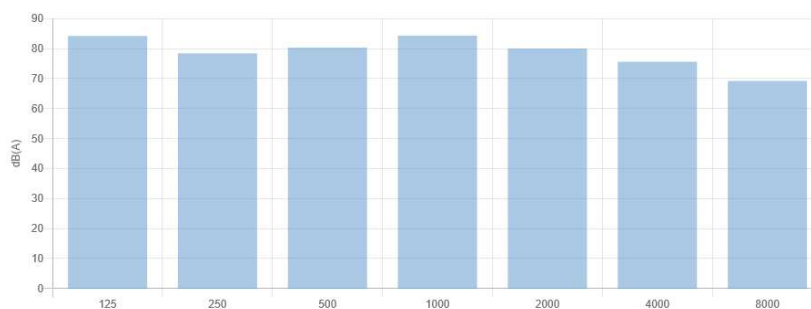
ACOUSTICS

www.hgcacoustics.com

**Sound data (nominal cooling data)**

Sound power - Lw	dB(A)	89.3
Sound pressure at 32.81 ft	dB(A)	56.1

Hz	Lw [dB]	Lw [dB(A)]
125	100.3	84.2
250	87.0	78.4
500	83.5	80.3
1000	84.3	84.3
2000	78.8	80.0
4000	74.6	75.6
8000	70.4	69.2



The sound levels are given at full load, without pumps (if available) and at nominal conditions (air temperature: 95.0 °F, water temperature (in/out): 54.0/44.0 °F).

Sound power: calculated on the basis of measurements carried out in accordance with the UNI EN ISO 9614-2 regulation, in compliance with the requirements of the Eurovent certification. Sound pressure calculated according to correlation in accordance with UNI EN ISO 3744.

Electric data

Locked Rotor Amps (LRA)	A	398.0
Minimum Circuit Ampacity (MCA)	A	600.0
Maximum Overcurrent Protection (MOP)	A	700.0
Short-Circuit Current Rating (SCCR)	kA	10
Power supply		575V/3/60Hz

Dimensions and weights

A - Height	in	100.39
B - Width	in	86.61
C - Length	in	468.5
Empty weight	lb	22,817.84
Working weight	lb	24,978.37
Shipping weight	lb	22,817.84

As specified in the conditions of use, the technical data shown are not binding; Aermec reserves the right to make changes for improvements or corrections at any time.

Appendix D

Road Traffic Data & Sample Calculations



NOISE



VIBRATION



ACOUSTICS

www.hgcacoustics.com

June 10, 2026

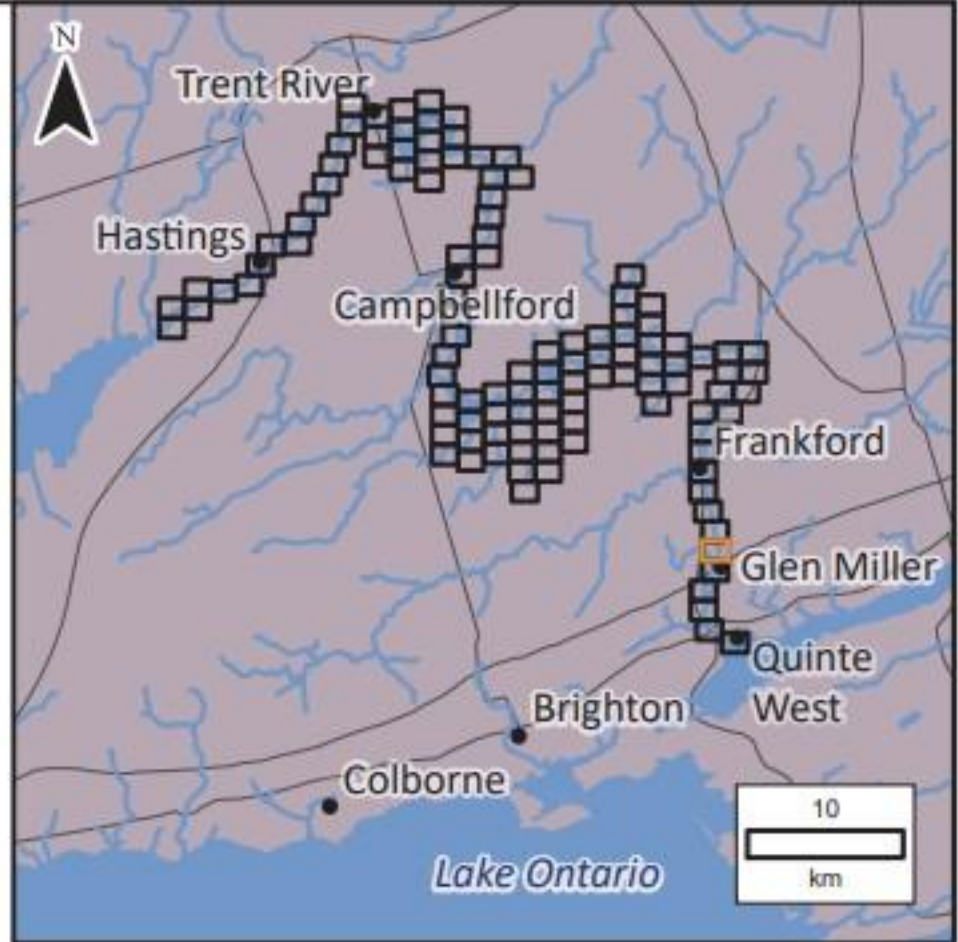
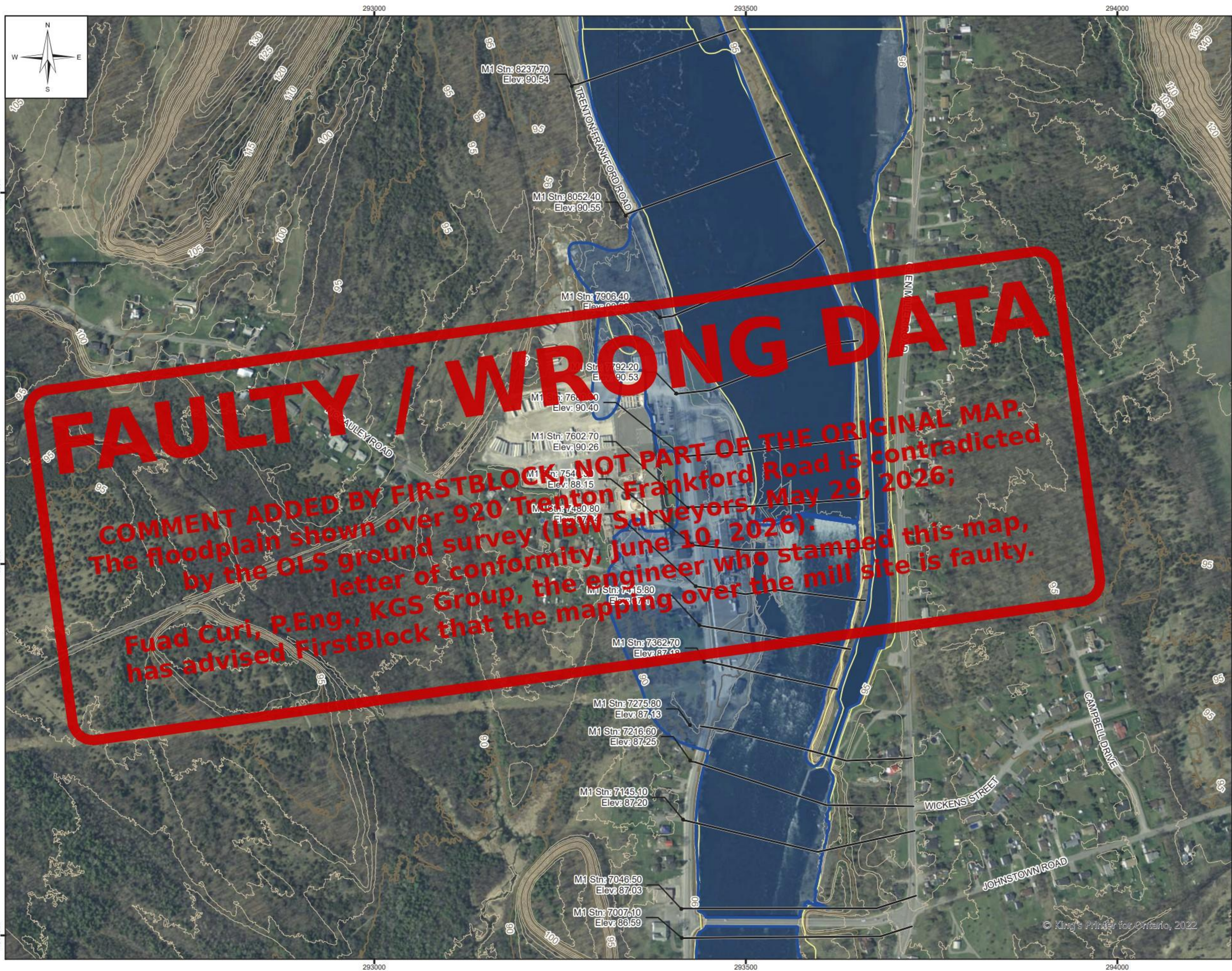
Doug Randell
doug@fusionconsultingltd.ca

Re: A-059675 - Partial Topo Base Plan with Boundary Linework - 920 Trenton Frankford Road, Trenton

Based on the topographic survey completed on June 3, 2026 by IBW Surveyors Ltd., all the elevations within the subject property limits shown on A-059675-TOPO_v4 are above the M1 Stn 7906.40 elevation 90.56M noted on the Regulatory Flood Mapping dated February 2024. The Regulatory Flood Mapping (February 2024) and topographic survey completed by IBW Surveyors Ltd. are attached for reference.



Cassandra Buma
Project Manager
cassandra.buma@ibwsurveyors.com



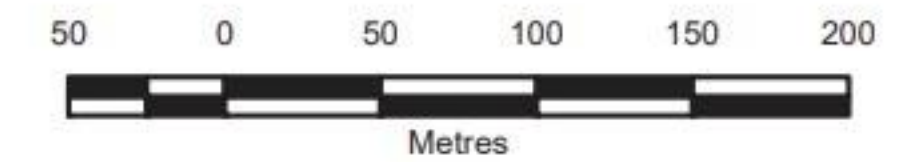
LEGEND:

- Station Number, Regulatory Flood Elevation (metres)
- 80.02 Water Surface Elevation
- Cross Section
- 5 m Index Contour
- 1 m Contour
- Existing Floodlines
- Regulatory Floodplain (100 Year - 1% AEP)



NOTES:

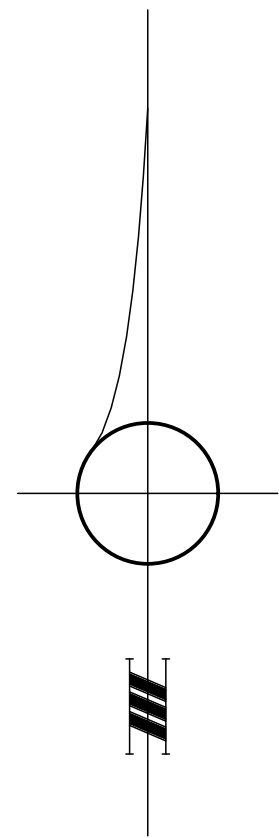
- LIDAR elevation dataset provided by Land Information Ontario (LIO) and Natural Resources Canada (NRCAN). Datasets included OMAFRA (Peterborough) dated 2016-2017 (LIO) and Belleville dated 2022 (NRCAN).
- Aerial photography, dated 2013, was provided by Land Information Ontario (LIO) as part of the South Central Orthophotography Project (SCOOP).
- Maps are prepared in Projection NAD 83, UTM Zone 18, CSRS. Vertical reference datum used is Canadian Geodetic Vertical Datum of 2013 (CGVD2013). Elevations are in metres above sea level (MSL).
- The Regulatory Floodplain lines shown in this map were prepared using hydraulic models as described in: KGS Group, 2024, "Trent River Floodplain Mapping Update - Floodplain Mapping Report"



SCALE: 1:5,000 METRIC 11"x17"

All units are metric and in metres unless otherwise specified. Transverse Mercator Projection, NAD83 UTM Zone 18 CSRS. Elevations are in metres above sea level (MSL).

0	24/02/29	ISSUED WITH FINAL REPORT	FGC	AB
NO.	YYMMDD	DESCRIPTION	ISSUED BY	CHECK BY
REVISIONS / ISSUE				
				
TRENT RIVER FLOODPLAIN MAPPING UPDATE				
REGULATORY FLOODPLAIN MAP				
FEBRUARY 2024		(SHEET 112 OF 117)	REV.	0



IMAGERY
AERIAL IMAGERY SHOWN IS FOR ILLUSTRATIVE PURPOSES ONLY AND
MAY NOT DEPICT CURRENT FEATURES.

COPYRIGHT © IBW SURVEYORS LTD. 2020
TOPOGRAPHIC BASE PLAN OF
920 TRENTON FRANKFORD ROAD
TOWNSHIP OF QUINTE WEST
SCALE 1:400 METRES

LEGEND	
	AC DENOTES AIR CONDITIONER UNIT
	AN DENOTES ANCHOR
	BF DENOTES BOARD FENCE
	BL DENOTES BOLLARD
	CB DENOTES CATCH BASIN
	CLF DENOTES CHAIN LINK FENCE
	CO DENOTES CLEAN-OUT
	CONC DENOTES CONCRETE
	UP DENOTES UTILITY POLE
	UPLS DENOTES UTILITY/LIGHT STANDARD
	FHY DENOTES FIRE HYDRANT
	DSE+ DENOTES DOOR SILL ELEVATION
	PH DENOTES PRE-HYDRANT
	GSE+ DENOTES GARAGE SILL ELEVATION
	INV DENOTES INVERT
	MH DENOTES MAINTENANCE HOLE
	MW DENOTES MONITORING WELL
	MB DENOTES MAILBOX
	SIG DENOTES SIGN
	WCAP DENOTES WELL CAP ELEVATION AT TOP CENTRE
	AC DENOTES WATER VALVE
	HW DENOTES HANDRAIL
	OVW DENOTES OVERHEAD WIRES
	CT DENOTES CONIFEROUS TREE W/ TRUNK DIAMETER
	DD DENOTES DECIDUOUS TREE W/ TRUNK DIAMETER

CAUTION
THIS IS NOT A PLAN OF SURVEY AND SHALL NOT BE USED EXCEPT
FOR THE PURPOSE INDICATED IN THE TITLE BLOCK.

COORDINATE NOTE
COORDINATE VALUES AND DIGITAL FILE ARE IN GRID SYSTEM, UTM
ZONE 18 (75° WEST LONGITUDE), NAD83(CSR85+7/2010).
COMBINED SCALE FACTOR = 1.000006

CONTOUR NOTE
CONTOURS SHOWN HEREON ARE DRAWN AT 0.20 METRE
INTERVALS.

ELEVATION NOTE
ELEVATIONS ARE GEODETIC AND REFERRED TO THE CANADIAN
GEODETIC VERTICAL DATUM (CGVD2011) BY DIRECT MEASUREMENT
TO A REAL TIME NETWORK.

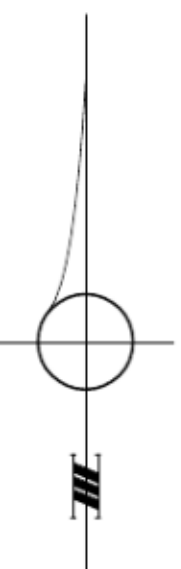
BOUNDARY NOTE
THE BOUNDARIES OF INTEREST WERE COMPILED FROM EXISTING
SURVEY AND TITLE RECORDS. FIELDWORK WAS COMPLETED TO
LOCATE SURVEY MONUMENTS FOR THE PURPOSES OF
GEOREFERENCING THE COMPILED BOUNDARY TO NAD83(CSR85+7/2010). UTM ZONE IS GRID COORDINATES. THE ACCURACY OF THE
COMPILED BOUNDARY IS NOT GUARANTEED, AND IS SUBJECT TO
ANY POSSIBLE ERRORS IN THE RECORDS REFERENCED. THE
COMPILED BOUNDARY IS INTENDED TO BE USED FOR CONCEPTUAL
DESIGN AND DISCUSSION PURPOSES ONLY. IBW SHALL NOT BE
RESPONSIBLE FOR THE USE OF, OR THE RELIANCE ON, THE
DELIVERABLES PROVIDED IN PLACE OF A LEGAL BOUNDARY SURVEY.



IBWSURVEYORS.COM | 1.800.667.0696

DATE: 2020-07-20 10:00:00 AM 2020-07-20 10:00:00 AM 2020-07-20 10:00:00 AM 2020-07-20 10:00:00 AM
A-059675

NOTICE: ALL BLUE AND RED MARKUPS ON THIS DRAWING WERE ADDED BY FIRSTBLOCK.
They are not part of the IBW Surveyors plan.



Lowest ground, north frontage
Ground (survey): 91.12 m
Flood elevation (7906.40): 90.56 m
Ground is 0.56 m above flood level

Lowest ground, yard west of north wing
Ground (survey): 91.30 m
Flood elevation (7792.20): 90.53 m
Ground is 0.77 m above flood level

RESULT: NO FLOOD ZONE ON THE PROPERTY

**No surveyed ground at 920 Trenton Frankford Road is below
the regulatory flood elevation at any station.
Lowest ground: 91.12 m. Highest flood elevation: 90.56 m.
Minimum clearance: 0.56 m. Blue flood area: none.**

Lowest ground, south corner
Ground (survey): 91.14 m
Flood elevation (7602.70): 90.26 m
Ground is 0.88 m above flood level

ESTIMATED FLOOD ZONE, 920 TRENTON FRANKFORD ROAD

- Flood zone: ground below the regulatory flood elevation at the nearest station. Result: no such area on the property.
- Containment wall along the road frontage, top elevation 94.1 m

Method: spot elevations from the IBW Surveyors topographic plan (May 29, 2026, datum CGVD28:78; pipe inverts excluded) interpolated to a ground surface. Regulatory flood elevations from the KGS Group / Lower Trent Conservation Trent River Floodplain Mapping Update (Feb. 2024, sheet 112, datum CGVD2013), interpolated between stations along the Trenton Frankford Road frontage (station positions approximate, north end at Stn 7906.40). Ground is compared point by point to the flood elevation at that location; dark blue would mark any area where the flood level is higher than the ground. Lowest surveyed ground on the property is 91.12 m; highest adjacent flood elevation is 90.56 m. The containment wall line is schematic, drawn along the station alignment. Clearances are before datum conversion. Prepared by FirstBlock for discussion; not a flood hazard determination; to be confirmed by the OLS.

M1 Stn 7906.40
regulatory flood elev. 90.56 m

M1 Stn 7792.20
regulatory flood elev. 90.53 m

**CONTAINMENT WALL, top elevation 94.1 m
3.54 m above the highest regulatory flood elevation (90.56 m)**

M1 Stn 7680.30
regulatory flood elev. 90.40 m

M1 Stn 7602.70
regulatory flood elev. 90.26 m

M1 Stn 7540.30
regulatory flood elev. 88.15 m

KEY PLAN: NOT TO SCALE

IMAGERY
AERIAL IMAGERY SHOWN IS FOR ILLUSTRATIVE PURPOSES ONLY AND MAY NOT DEPICT CURRENT FEATURES.
COPYRIGHT © IBW SURVEYORS LTD. 2026
TOPOGRAPHIC BASE PLAN OF
**920 TRENTON FRANKFORD ROAD
TOWNSHIP OF QUINTE WEST**
SCALE: 1:400 METRES

LEGEND

AC	DENOTES AIR CONDITIONER UNIT
AN	DENOTES ANCHOR
BF	DENOTES BOARD FENCE
BL	DENOTES BOLLARD
CB	DENOTES CATCH BASIN
CLF	DENOTES CHAIN LINK FENCE
CO	DENOTES CLEAN-OUT
CONC	DENOTES CONCRETE
CU	DENOTES UTILITY CULVERT
ULS	DENOTES UTILITY LIGHT STANDARD
DOOR	DENOTES DOOR SILL ELEVATION
FRH	DENOTES FIRE HYDRANT
GSE	DENOTES GARAGE SILL ELEVATION
MAH	DENOTES MAINTENANCE HOLE
MHW	DENOTES MONITORING WELL
MB	DENOTES MAILBOX
SD	DENOTES SIGN
WCE	DENOTES WELL CAP ELEVATION AT TOP CENTRE
WV	DENOTES WATER VALVE
HW	DENOTES HANDRAIL
OW	DENOTES OVERHEAD WIRES
CTD	DENOTES CONIFEROUS TREE WITH TRUNK DIAMETER
DD	DENOTES DECIDUOUS TREE WITH TRUNK DIAMETER

CAUTION
THIS IS NOT A PLAN OF SURVEY AND SHALL NOT BE USED EXCEPT FOR THE PURPOSE INDICATED IN THE TITLE BLOCK.

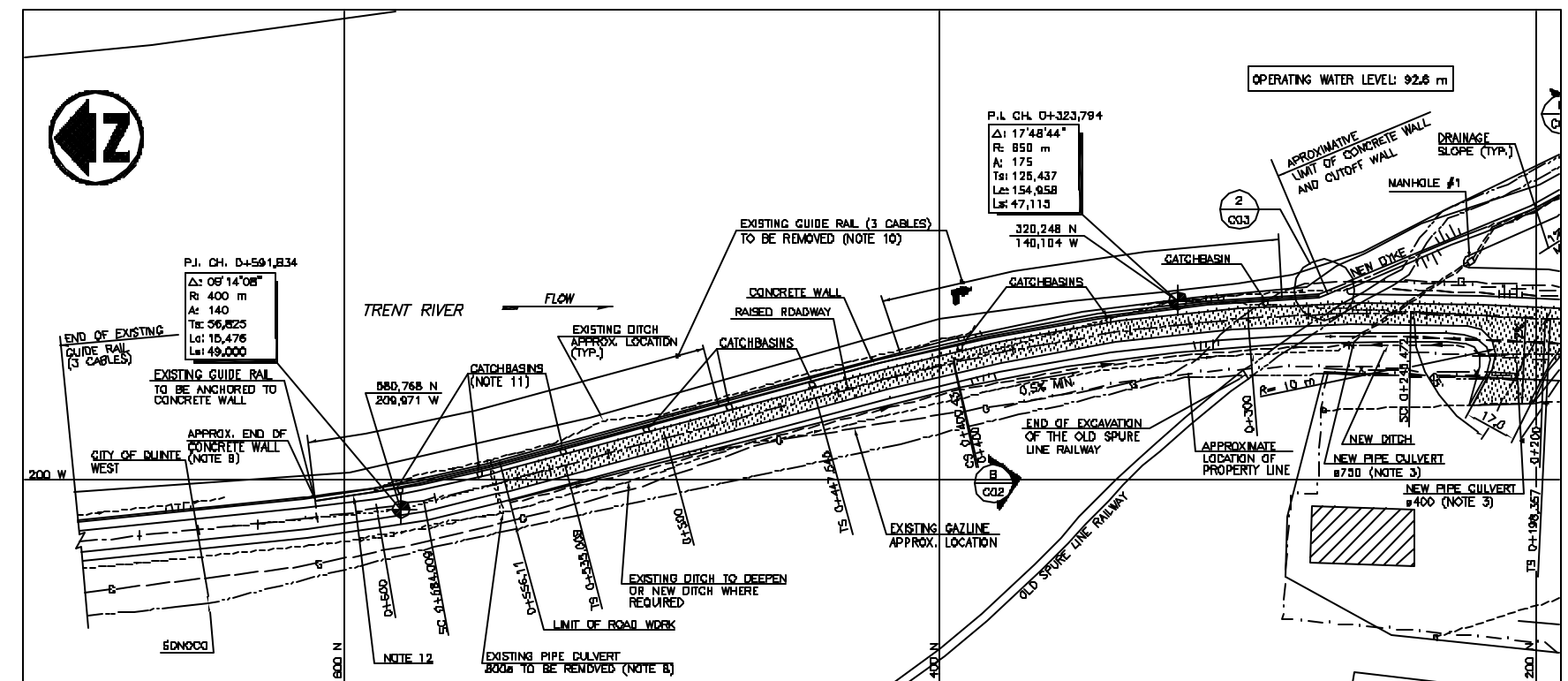
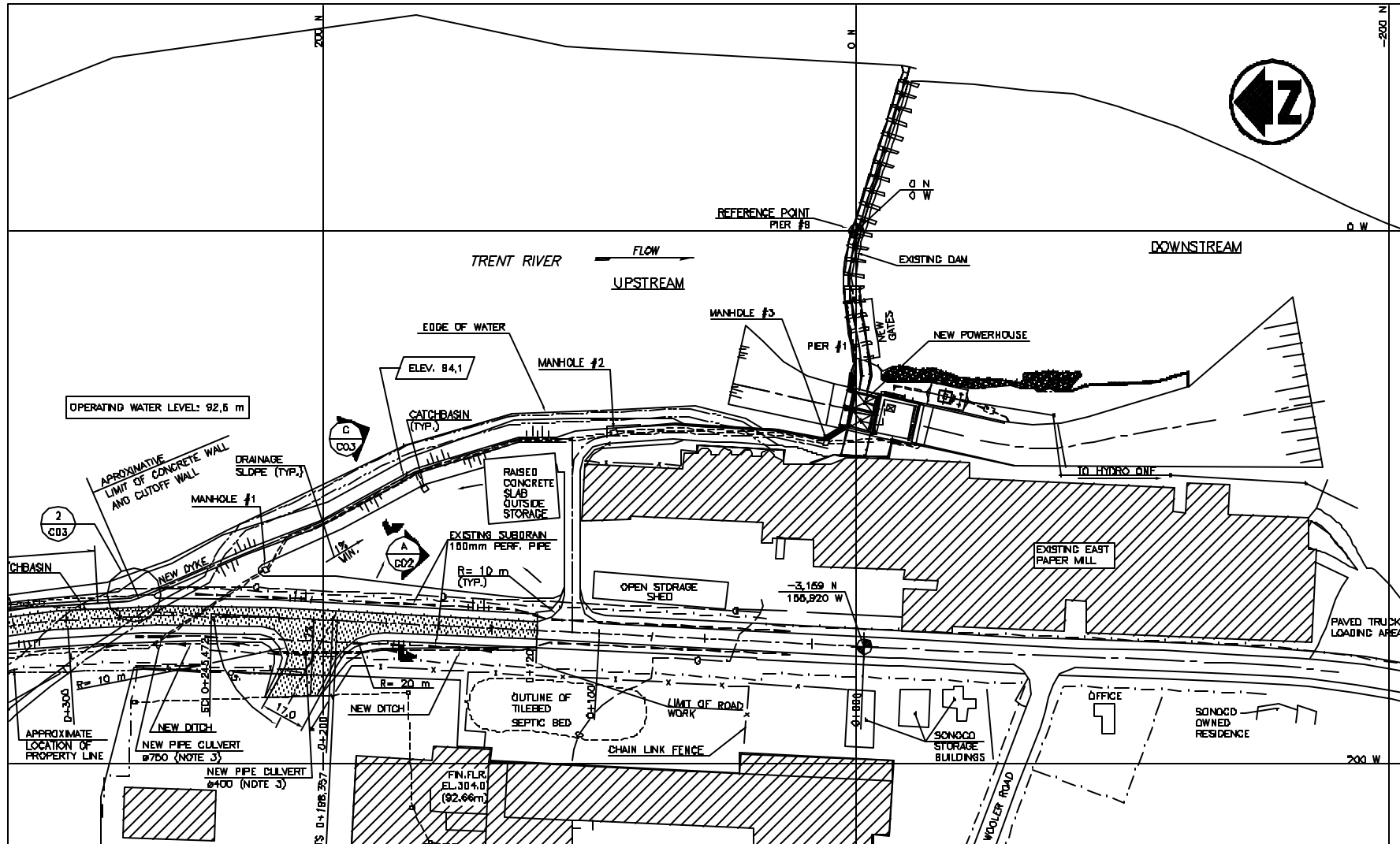
COORDINATE NOTE
COORDINATE VALUES AND DIGITAL FILE ARE IN GRID SYSTEM UTM ZONE 18 T (77 WEST LONGITUDE NAD83(CRS)10200). COMBINED SCALE FACTOR = 1.00016

CONTOUR NOTE
CONTOURS SHOWN HEREON ARE DRAWN AT 0.20 METRE INTERVALS.

ELEVATION NOTE
ELEVATIONS ARE GEODETIC AND REFERRED TO THE CANADIAN GEODETIC VERTICAL DATUM (CGVD2013) BY DIRECT MEASUREMENT TO A REAL-TIME NETWORK.

BOUNDARY NOTE
THE BOUNDARIES OF INTEREST WERE COMPILED FROM EXISTING SURVEY AND TITLE RECORDS. FIELDWORK WAS COMPLETED TO LOCATE SURVEY MONUMENTS FOR THE PURPOSES OF GEOREFERENCING THE COMPILED BOUNDARY TO NAD83(CRS)10200. UTM ZONE 18 T IS GRID COORDINATES. THE ACCURACY OF THE COMPILED BOUNDARY IS NOT GUARANTEED AND IS SUBJECT TO ANY POSSIBLE ERRORS IN THE RECORDS REFERENCED. THE COMPILED BOUNDARY IS INTENDED TO BE USED FOR CONCEPTUAL DESIGN AND DISCUSSION PURPOSES ONLY. IBW SHALL NOT BE RESPONSIBLE FOR THE USE OF OR THE RELIANCE ON THE BOUNDARIES PROVIDED IN PLACE OF A LEGAL BOUNDARY SURVEY.

IBWSURVEYORS.COM | 1.800.667.0596
A-05967



- Notes or references**
- ROAD CRITERIA CLASSIFICATION = RAISED.
MAX = 0.06 m/m.
GEOMETRIC DESIGN CRITERIA ARE IN ACCORDANCE WITH GEOMETRIC DESIGN GUIDE FOR CANADIAN ROADS EDITED BY TRANSPORTATION ASSOCIATION OF CANADA, SEPTEMBER 1996.
 - ALL WORK SHALL BE IN ACCORDANCE WITH OPSB AND OPSD.
 - POLYETHYLENE PLASTIC PIPE IN ACCORDANCE WITH OPSB 421 AND OPSD-802.014. RIRAP IS REQUIRED AT INLET IN ACCORDANCE TO OPSD 0810.0200.
 - PAVEMENT WIDENING ON BOTH SIDES OF CURVES IN ACCORDANCE WITH OPSD-213.01. FULL WIDENING OF 0.9 m.
 - SUPER ELEVATION: 5.1% FOR R= 400.0
4.1% FOR R= 800.0
SLOPE 1: 400 FOR DEVELOPMENT FOR TANGENT RUNOUT
1: 200 FOR TRANSITION LENGTH.
 - THE SUPER ELEVATION SHALL MATCH THE EXISTING AT THE LIMIT OF WORK (D+120 AND D+350, 11).
 - REFERENCE DRAWINGS: MINISTRY OF TRANSPORTATION AND COMMUNICATION, ONTARIO CONT NO 80-18, NP NO 134-77-01, PLATES 13-33/8-0 AND 13-33/10-0.
 - THE LIMIT OF PROPERTIES ARE APPROXIMATE, RELATED FROM THE PLAN 21M OF P.A. MILLER SURVEYING LTD, DATED SEPTEMBER 28, 2003. THE EXACT LIMIT WILL BE SURVEY BEFORE ROAD CONSTRUCTION AS REQUIRED.
 - EXISTING CULVERT IF GOOD CONDITION HAULED TO THE FRANKFORD YARD. OTHERWISE IT IS TO BE DISPOSED OF BY THE CONTRACTOR. EXCAVATION TO BE BACK FILLED WITH SUITABLE EMPT FILL.
 - THE END OF CONCRETE WALL IS WHERE THE LEVEL OF THE RIDGE OF THE ROAD REACHES 94.1 m. AT THAT LOCATION CONCRETE WALL RISES 0.89 m ABOVE ROAD.
 - EXISTING GUIDE RAIL AND ANCHOR BLOCKS SHALL BE REMOVED AND DISPOSED OF BY THE CONTRACTOR.
 - THE OUTLET PIPE OF THE CATCHBASIN ALONG THE CONCRETE WALL IS CONNECTED TO THE 250 # HOPE PERFORATED PIPE LOCATED AT THE BASE OF THE WALL. FRAMES AND GRATES SHALL CONFORM TO OPSB 1830 AND OPSD 400.010. INSTALLATION IN ACCORDANCE TO OPSB 107. CATCHBASINS SHALL BE IN ACCORDANCE TO OPSD 706.010 (PRECAST CONCRETE 400 mm x 400 mm).
 - THE SHOULDER ADJACENT TO THE CONCRETE WALL BETWEEN THE LIMIT OF ROAD WORK (D+350, 11) AND THE END OF CONCRETE WALL (+/- 0+500) SHALL BE PAVED TO MATCH EXISTING PAVEMENT DEPTH.

N°	Date	Issues and/or revisions	Approved	Approved
2	2004-06-23	GENERAL REVISION		
1	2004-06-26	GENERAL REVISION		
0	2004-02-27	ISSUED FOR CONSTRUCTION		



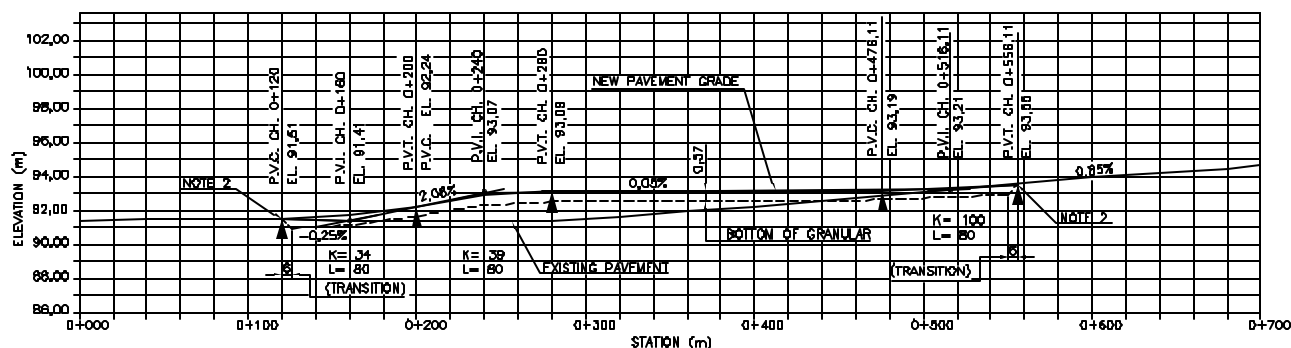
GLEN MILLER POWER INC.

1111, rue St-Charles Ouest
East Tower, Suite 1205
Longueuil, Québec J4K 3G4
Tel: (450) 826-2500
Fax: (450) 826-2544

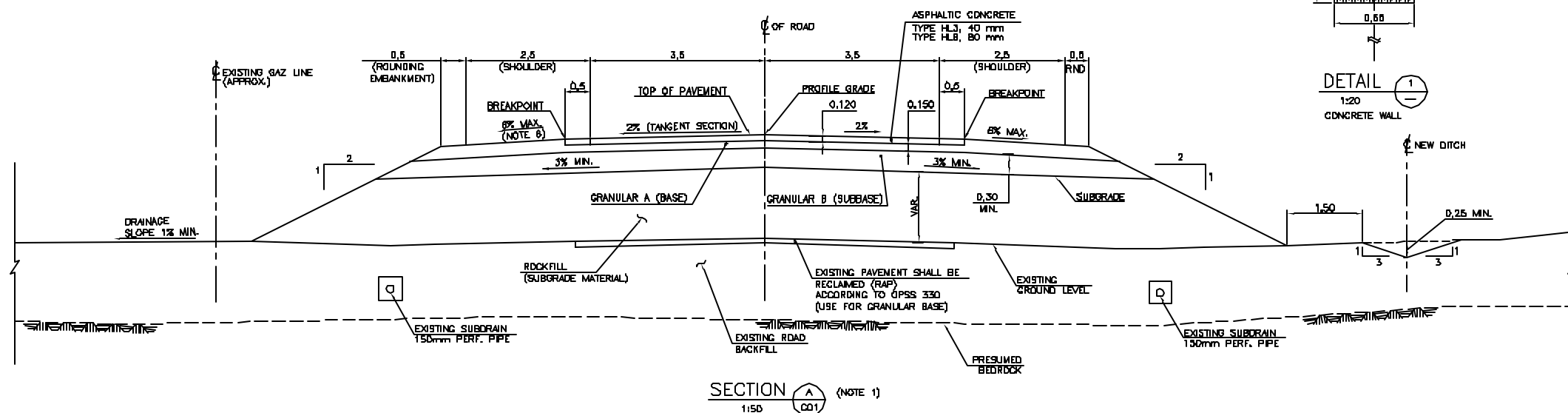
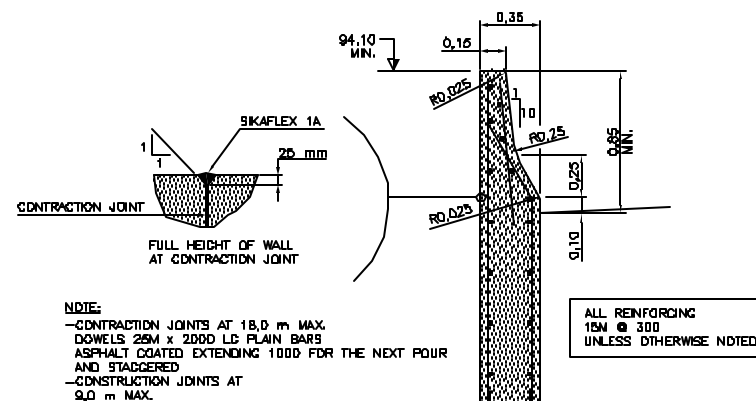


800, boul. René-Lévesque ouest, bureau 2600
Montréal (Québec) Canada H3B 1Z1
Téléphone: (514) 678-2021
Télécopieur: (514) 367-0065
courriel: rnav@glm-inc.com

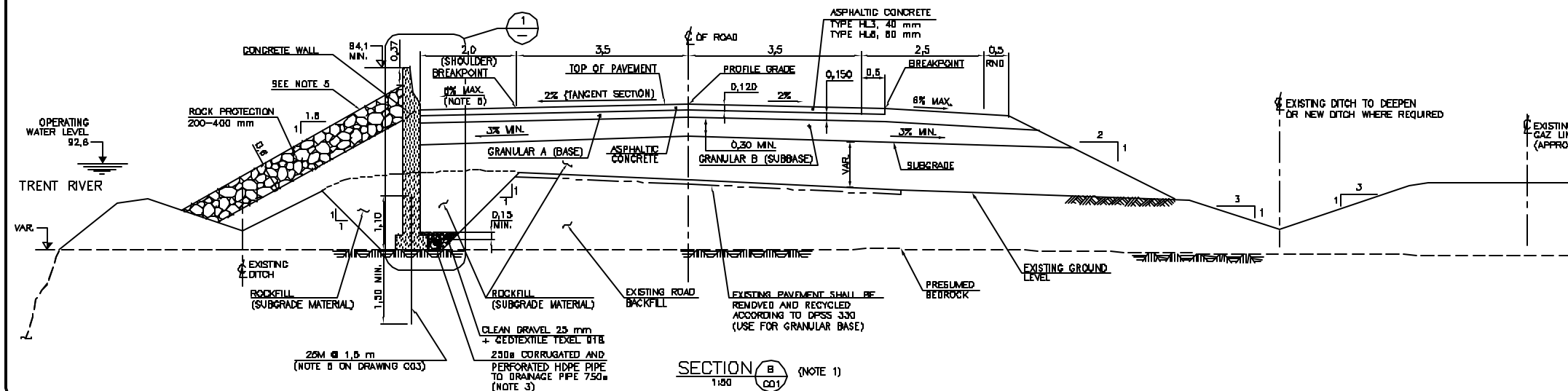
Project:	GLEN MILLER HYDROELECTRIC PROJECT			
Title:	FRANKFORD - STIRLING ROAD (HIGHWAY 33) RAISING WORKS PLAN VIEW			
Drawn:	G. NOREAU	Checked:	R. QUENNEVILLE	
Designed:	N. BEAUSÉJOUR	Checked:	S. GAGNER	
Approved:	C. CHARTRAND	Approved:	J. HANSON	
Scale factor 1:1000	Scale 0 10.0 50.0	Project n°	P48-0286	
Dimensions in m				
Customer:	Division:	Project:	Drawing n°	Mod. 2



VERTICAL PROFILE
HORIZ.: 1:2000
VERT.: 1:200



SECTION A-A (NOTE 1)
1:50



SECTION B-B (NOTE 1)
1:50

- Notes or references
- 1 THE SUBGRADE, SUBBASE AND BASE MATERIALS IN ACCORDANCE WITH OPS 314, OPS 321 AND OPS 1012.
 - 2 THE ASPHALTIC CONCRETE IN ACCORDANCE WITH OPS 310 AND OPS 1180.
 - 3 TRANSITION LAP JOINT DETAIL FOR PAVING IN ACCORDANCE WITH OPS 507.010.
 - 4 THE 250 & PIPE SHALL BE SHIFTED WHERE THE CATCHBASINS ARE LOCATED. THE CLEAN GRAVEL (25 mm) SHALL BE PLACED 0.15 m MIN. ALL AROUND THE PIPE.
 - 5 CONCRETE 35MPa, CLASS F-1, 5-8% ENTRAINED AIR.
 - 6 WHEN ROCK PROTECTION FILL COMES IN CONTACT DIRECTLY WITH RIVER OR SHORE, ROCK FILL SHALL BE CLEAN AND FREE OF FINES.
 - 7 SHOULDER DRESS FALL IN ACCORDANCE WITH OPSD-21002.

N°	Date	Issues and/or revisions	Approved	Approved
2	2004-06-06	GENERAL REVISION		
1	2004-06-25	GENERAL REVISION		
0	2004-02-27	ISSUED FOR CONSTRUCTION		



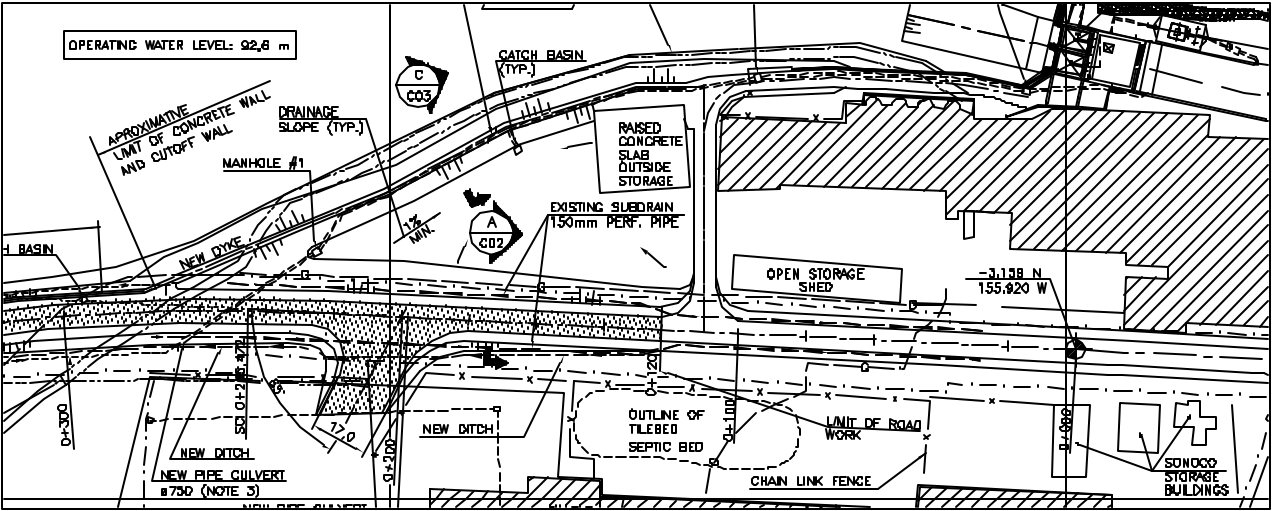
INNERGEX

1111, rue St-Charles Ouest
East Tower, Suite 1255
Longueuil, Québec J4K 3G4
Tel: (438) 826-2500

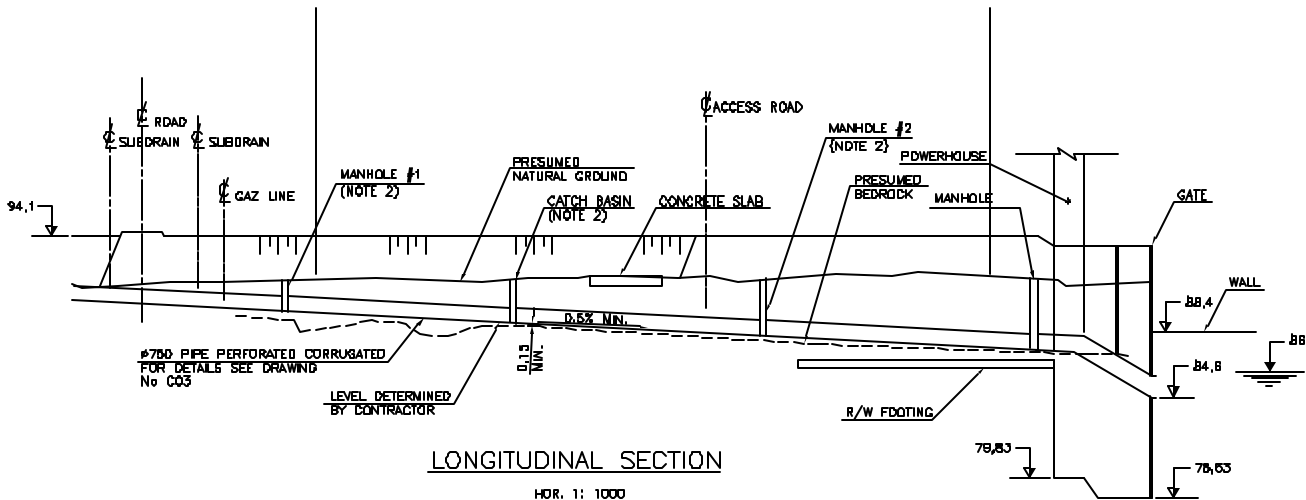


800, boul. René-Lévesque ouest, bureau 2600
Montréal (Québec) Canada H3B 1Z1
Téléphone: (514) 678-2021
Télécopieur: (514) 387-0085
courriel: rnv@rnfinc.com

Project	SONOCO HYDROPOWER EXTENSION			
Title	FRANKFORD - STIRLING ROAD (HIGHWAY 33) RAISING WORKS PROFILE AND SECTIONS			
Drawn	G. NOREAU	Checked	R. QUENNEVILLE	
Designed	N. BEAUSÉJOUR	Checked	S. GAGNIER	
Approved	C. CHARTRAND	Approved	J. HANSON	
Scale factor	1:2000	Scale	0 20 100	Project n° P48-0286
Dimensions in m				
Customer	P48	Division	0286	Mod. 2
		Project	C02	Rev.



PLAN



LONGITUDINAL SECTION

HOR. 1: 1000
VER. 1: 200

N°	Notes or references
1	CULVERT DESIGN CRITERIA: - DRAINAGE AREA: 27 ha - FLOW: 0.8 m³/sec
2	MANHOLE: 1800 mm DIAMETER PRECAST CONCRETE MAINTENANCE HOLE, MANUFACTURED IN ACCORDANCE WITH QPSD 701.040 AND INSTALLED IN ACCORDANCE WITH QPSD 407.

N°	Date	Issues and/or revisions	Approved	Approved
1	2004-08-26	GENERAL REVISION		
2	2004-09-27	ISSUED FOR CONSTRUCTION		



INNERGEX

1111, rue St-Charles Ouest
East Tower, Suite 1205
Longueuil, Québec J4K 5K4
Tel: (438) 828-2330
Fax: (450) 828-2544

REVIEWS

800, boulevard Lavoie ouest, bureau 2600
Montreal (Québec) Canada H3B 1Z1
Téléphones: (514) 878-2021
télécopieur: (514) 387-0085
courriel: rev@revinc.com

conseillers en ingénierie * engineering consultants

Project: SONOCO HYDROPOWER EXTENSION

Title: FRANKFORD - DRAINAGE PIPE RAISING WORKS
PLAN AND SECTIONS

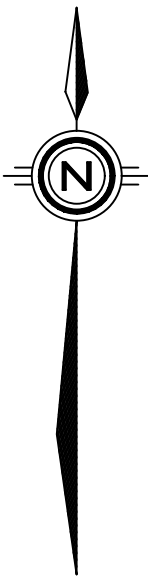
Drawn: G. NOREAU Checked: R. QUENNEVILLE
Designed: N. BEAUSÉJOUR Checked: S. GADNER
Approved: C. CHARTRAND Approved: J. HANSON

Scale factor 1:1000 Scale: 0 10 60 Project no: P48-0286

Dimensions in m

P48	0286	C05	Mod. 1
Customer	division	Project	Drawing no

FRANKFORD DATA CENTRE QUINTE WEST, ON



INDEX OF DRAWINGS

- C000 TITLE SHEET
- C001 GENERAL NOTES
- C002 KEY PLAN
- C003 DEMOLITION PLAN
- C004 SITE PLAN
- C005 GRADING PLAN
- C006 GRADING SECTIONS
- C007 TRENTON FRANKFORD ROAD SECTIONS
- C008 TRENTON FRANKFORD ROAD SECTIONS



DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	PRELIMINARY - NOT FOR CONSTRUCTION	DB	DSR					

THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION.
THESE DRAWINGS SHOULD NOT BE SCALED. IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

SCALE AS NOTED
DATE -
APPROVED DSR
DESIGN BY DSR
DRAWN BY DB
CHECKED BY DSR



FUSION

PROJECT MANAGEMENT | CIVIL ENGINEERING

FRANKFORD DATA CENTRE, QUINTE WEST, ON
TITLE PAGE

REV. No.	A
PROJECT No.	2026-019
DRAWING No.	C000

GENERAL NOTES

1. THE CONTRACTOR SHALL CONTACT THE OWNER TO ARRANGE A PRE-CONSTRUCTION MEETING PRIOR TO THE START OF CONSTRUCTION.
2. CONTACT "BC ONE CALL" AT 1-800-474-6886 A MINIMUM OF 72 HOURS PRIOR TO CONSTRUCTION TO LOCATE ALL EXISTING UTILITIES.
3. ALL CONSTRUCTION AND MATERIALS SHALL BE IN ACCORDANCE WITH:

- APPLICABLE LOCAL MUNICIPAL BYLAWS AND GUIDELINES.

- THE MASTER MUNICIPAL CONSTRUCTION DOCUMENT AND STANDARD DETAIL DRAWINGS (MMCD) LATEST EDITION (PLATINUM EDITION);

- CONSTRUCTION PLANS, SEALED AND ISSUED FOR CONSTRUCTION

- APPLICABLE CONTRACT DOCUMENTS AND ALL SPECIFICATIONS REFERENCED THEREIN.

- FLNED AND/OR MINISTRY OF ENVIRONMENT AND/OR FEDERAL DEPARTMENT OF FISHERIES AND OCEANS REQUIREMENTS;

- MINISTRY OF TRANSPORTATION "B. C. TRAFFIC CONTROL MANUAL FOR WORK ON ROADWAYS", LATEST EDITION;

- WORKSAFE BC LATEST EDITION.

- HEALTH AUTHORITY.
- THE CONTRACTOR SHALL MAINTAIN ON SITE COPIES OF THE ABOVE DOCUMENTS AND SHALL ENSURE THAT ALL TRADES ARE THOROUGHLY FAMILIAR WITH THE APPLICABLE SECTIONS OF THE DOCUMENTS.
4. THE CONTRACTOR SHALL EXPOSE AND VERIFY THE LOCATION AND ELEVATION OF ALL EXISTING SERVICES IN THE FIELD PRIOR TO CONSTRUCTION AND NOTIFY THE ENGINEER OF RECORD OF ANY DISCREPANCIES, CONFLICTS OR OMISSIONS PRIOR TO CONSTRUCTION.
5. FIGURED DIMENSION SHALL GOVERN OVER SCALED DIMENSIONS.
6. LOT DIMENSIONS SHOWN ARE FOR REFERENCE ONLY. REFER TO BCLS REGISTERED LEGAL PLAN FOR ACTUAL LOT DIMENSIONS. THE CONTRACTOR SHALL CONFIRM LOCATIONS OF LOT LINES PRIOR TO INSTALLATION OF SERVICE CONNECTIONS.
7. THE CONTRACTOR SHALL ENSURE THAT ALL APPROVALS REQUIRED FOR THE PROPOSED WORKS HAVE BEEN OBTAINED FROM ALL AUTHORITIES AND AGENCIES PRIOR TO COMMENCING THE WORK.
8. THE CONTRACTOR SHALL ARRANGE FOR AND COORDINATE THE WORKS DONE BY

- BC HYDRO, TELUS, CABLE AND GAS.

- OTHER FRANCHISE UTILITIES.
9. THE CONTRACTOR SHALL CONTACT THE APPROPRIATE PERSONNEL AT LEAST 72 HOURS PRIOR TO THE WORK. SCHEDULING AND OTHER CONSTRUCTION CONSTRAINTS IMPOSED BY THESE WORKS SHALL BE TAKEN INTO ACCOUNT.
10. THE CONTRACTOR SHALL REPAIR OR REPLACE ANY EXISTING STREETS, SERVICES, OR LANDSCAPING THAT MAY BE DAMAGED AS A RESULT OF CONSTRUCTION. REPAIRS TO EXISTING SERVICES SUCH AS WATER, SANITARY SEWER, STORM SEWER AND STREET OR TRAFFIC LIGHTING SHALL BE MADE BY THE CITY AT COST TO THE CONTRACTOR UNLESS OTHERWISE AGREED TO. REPAIRS TO EXISTING SURFACE WORKS MAY BE DONE BY THE CONTRACTOR AT THE DISCRETION OF THE CITY.
11. THE CONTRACTOR SHALL ENSURE THAT STREETS ARE KEPT CLEAN AND FREE OF EQUIPMENT AND MATERIALS AT ALL TIMES WHEN CONSTRUCTION ACTIVITY IS NOT UNDERWAY.
12. A TRAFFIC AND PEDESTRIAN SAFETY CONTROL PLAN SHALL BE SUBMITTED BY THE CONTRACTOR PRIOR TO THE PRE-CONSTRUCTION MEETING.
13. CONTRACTOR SHALL BE REGISTERED WITH WORKSAFE BC.

EARTHWORKS & GRADING

1. THE CONTRACTOR SHALL ENSURE ALL CONSTRUCTION ACTIVITIES ARE IN ACCORDANCE WITH THE GEOTECHNICAL RECOMMENDATIONS. WHERE THE GEOTECHNICAL REPORT VARIES FROM THE STANDARD MMCD 2009 (PLATINUM) EDN. SECTION 31 CLAUSES, THE REQUIREMENTS OF THE GEOTECHNICAL REPORT SHALL TAKE PRECEDENCE.
2. ALL TESTING OF MATERIALS AND COMPACTION SHALL BE PERFORMED BY A GEOTECHNICAL ENGINEER. PRIOR TO ISSUANCE OF THE SUBSTANTIAL COMPLETION CERTIFICATE, THE SUPERVISING CONSULTING ENGINEER SHALL SUBMIT TO THE CITY A FINAL GEOTECHNICAL REPORT PREPARED BY THE GEOTECHNICAL ENGINEER. COPIES OF REPORTS AND TEST RESULTS SHALL BE SUBMITTED TO THE ENGINEER OF RECORD AS WORK PROGRESSES OR AS REQUESTED.
3. GEOTECHNICAL ENGINEER TO REVIEW AND APPROVE EXISTING MATERIAL, TRENCH BACKFILL, AND IMPORTED MATERIAL PRIOR TO PLACEMENT.
4. REMOVE VEGETATION, STRIP ORGANIC MATERIAL, AND UNSUITABLE MATERIAL, AS REQUIRED, PRIOR TO PLACING STRUCTURAL FILL.
5. COMPACT ALL EARTH FILLS AND GRANULAR MATERIAL LAYERS TO 95% MODIFIED PROCTOR DENSITY (MPD), UNLESS OTHERWISE DIRECTED BY THE GEOTECHNICAL ENGINEER.
6. AREAS SHALL BE GRADED TO +/- 50mm AT NOTED CONTROL POINTS. FINAL GRADING SHALL BE COMPLETED PRIOR TO THE ISSUANCE OF A SUBSTANTIAL COMPLETION CERTIFICATE.
7. EMBANKMENTS SHALL BE PLACED, SHAPED, COMPACTED, AND PROTECTED TO STANDARDS AND SPECIFICATIONS OF THE GEOTECHNICAL ENGINEER.
8. ALL SIEVE, PROCTOR AND DENSITY TEST RESULTS TO BE PROVIDED TO THE ENGINEER OF RECORD.

EROSION AND SEDIMENT CONTROL

1. THE CONTRACTOR IS RESPONSIBLE FOR ENSURING THAT BMPs ARE IMPLEMENTED ON SITE. ACCEPTABLE BMPs INCLUDE THOSE PROVIDED BELOW, THOSE PROVIDED BY THE EROSION AND SEDIMENT CONTROL SUPERVISORS, AND THOSE PROVIDED IN FISHERIES AND OCEANS CANADA'S LAND DEVELOPMENT GUIDELINES FOR THE PROTECTION OF AQUATIC HABITAT (1993):

a. CLEAR ONLY THE AREAS NECESSARY (LEAVE VEGETATION IN AREAS THAT DON'T REQUIRE DISTURBANCE)

b. ESTABLISH A STABILIZED ENTRY/EXIT POINT

c. PROTECT THE PERIMETER OF THE SITE

d. DIVERT UP-SLOPE WATER AROUND THE WORK SITE (KEEP CLEAN WATER CLEAN)

e. ROUGHEN EXPOSED SOILS

f. TEMPORARILY STABILIZE DISTURBED EARTH

g. CREATE TEMPORARY SEDIMENT DETENTION FACILITIES ONSITE

h. INSTALL INLET PROTECTION MEASURES AT ALL STORM SYSTEM INLETS

i. PLACE SOIL PILES UPSLOPE OF THE PERIMETER PROTECTION AND COVER WITH PLASTIC SHEETING OR EROSION CONTROL BLANKETS

j. DEWATER TO A CONTAINED PART OF THE SITE AND ALLOW IT TO INFILTRATE INTO THE SOIL

k. INSTALL ROOF DOWNPIPES AS SOON AS PRACTICABLE AFTER THE ROOF IS LAID

l. ENSURE THAT ALL CONTROL MEASURES ARE MAINTAINED IN GOOD WORKING ORDER

m. RE-VEGETATE OR OTHERWISE PERMANENTLY STABILIZE THE SITE
2. THE CONTRACTOR SHALL INSPECT, DOCUMENT, AND MAINTAIN ESC WORKS REGULARLY, AND AFTER EACH SIGNIFICANT RAINFALL EVENT. RECOMMENDATIONS CONCERNING ADDITIONS, MAINTENANCE, AND MODIFICATIONS TO THE ACCEPTED PLAN, OR NEW BEST MANAGEMENT PRACTICES (BMPs), SHALL BE IMPLEMENTED IMMEDIATELY BY THE CONTRACTOR.
3. THE CONTRACTOR IS RESPONSIBLE FOR THE MODIFICATION AND MAINTENANCE OF THE ESC WORKS ON AN ONGOING BASIS, AND IMMEDIATELY PERFORM REMEDIAL WORKS AS DIRECTED BY THE ESCs, THE MUNICIPALITY HAVING JURISDICTION, ENGINEER OF RECORD, OR OTHER REGULATORY AND GOVERNMENT AGENCIES.

ROADWORKS

1. CONTRACTOR TO ARRANGE FOR REQUIRED TESTING AT LEAST 48 HOURS PRIOR TO COVERING WORKS.
2. CHANGES OF GRADE AND DIRECTION SHALL BE FORMED BY SMOOTH VERTICAL AND HORIZONTAL CURVES.
3. ALL SUB-BASE ROAD AND GRANULAR BASE MATERIALS SHALL BE COMPACTED TO 95% MPD.
4. EARTHWORKS, INCLUDING STRIPPING ALL LOOSE OR OTHERWISE DELETERIOUS MATERIALS OR SOFT SPOTS, COMPACTING, AND SLOPE GRADING, SHALL BE COMPLETED TO THE APPROVAL OF A GEOTECHNICAL ENGINEER.
5. THE CONDITIONS FOR PLACING ASPHALT PAVEMENT AND CONCRETE SHALL CONFORM WITH THE SPECIFICATIONS DETAILED IN MMCD 2009 (PLATINUM) EDN.
6. TIE-INS TO EXISTING PAVEMENT SHALL BE MADE BY CUTTING BACK THE EXISTING PAVEMENT TO SOUND MATERIAL AS NECESSARY TO PRODUCE A NEAT VERTICAL FACE. PRIOR TO PLACING ASPHALTIC CONCRETE, EXPOSED PAVEMENT FACES AND OTHER ABUTTING STRUCTURES SHALL BE TACK COATED WITH ASPHALT EMULSION.
7. ALL NEW PAVEMENT SHALL BE GRADED SO THAT NO PONDING FORMS. THE MINIMUM CROSS SLOPE ON ANY NEW OR OVERLAIN ROAD SURFACE SHALL BE 2%, UNLESS OTHERWISE SPECIFIED.
8. ALL VALVES BOXES, MANHOLES, ETC., THAT ARE WITHIN THE PAVED SURFACE OF THE ROADWAY SHALL BE SET AT BASE COURSE ELEVATION FOR THE WARRANTY PERIOD, AND ADJUSTED TO FINISHED COURSE ELEVATION PRIOR TO FINISH COURSE PAVING, IF APPLICABLE.

SANITARY SEWER

1. TIE-INS AND CONNECTIONS TO EXISTING SANITARY SEWERS TO BE PERFORMED BY THE CONTRACTOR UNDER THE DIRECT SUPERVISION OF THE ENGINEER OF RECORD. A MINIMUM 72 HOURS NOTICE TO BE GIVEN FOR ANY TIE-IN.
2. NEW SEWER LINES TIED TO EXISTING LINES SHALL BE PLUGGED UNTIL THEY ARE TESTED AND FLUSHED.
3. POWER FLUSHING SHALL BE PERFORMED ON ALL MAINS PRIOR TO ACCEPTANCE. VIDEO TAPE OF ALL MAINS SHALL BE REVIEWED BY THE ENGINEER PRIOR TO PLACING IN SERVICE. TESTING AND VIDEO INSPECTION SHALL BE PERFORMED UNDER THE DIRECT SUPERVISION OF THE SUPERVISING CONSULTING ENGINEER. VIDEO TAPES AND TEST RESULTS SHALL BE PROVIDED TO THE ENGINEER PRIOR TO THE ISSUANCE OF A SUBSTANTIAL COMPLETION CERTIFICATE.
4. DELETERIOUS MATERIALS SHALL BE PREVENTED FROM ENTERING THE SEWER SYSTEM.
5. TOP OF INSPECTION CHAMBER STANDPIPES SHALL BE SET TO 600mm ABOVE FINAL LOT GRADE.
6. MANUFACTURED WYES OR APPROVED ALTERNATIVE SHALL BE USED FOR ALL SERVICE CONNECTIONS TO SANITARY MAINS.
7. PROVIDE A MINIMUM 1.5m COVER FOR SANITARY SEWERS.
8. ALL PIPE TO BE PVC DR35 OR AS NOTED.

STORM

1. TIE-INS AND CONNECTIONS TO EXISTING STORM SEWERS TO BE PERFORMED BY THE CONTRACTOR UNDER THE DIRECT SUPERVISION OF THE ENGINEER OF RECORD, FOLLOWING APPROVAL TO PROCEED FROM THIS CITY. A MINIMUM 72 HOURS NOTICE TO BE GIVEN FOR ANY TIE-IN.
2. NEW SEWER LINES TIED TO EXISTING LINES MUST BE PLUGGED UNTIL THEY ARE TESTED AND FLUSHED.
3. POWER FLUSHING SHALL BE PERFORMED ON ALL MAINS PRIOR TO ACCEPTANCE. VIDEO TAPE OF ALL MAINS SHALL BE REVIEWED BY THE ENGINEER PRIOR TO PLACING IN SERVICE. TESTING AND VIDEO INSPECTION SHALL BE PERFORMED UNDER THE DIRECT SUPERVISION OF THE SUPERVISING CONSULTING ENGINEER. VIDEO TAPES AND TEST RESULTS SHALL BE PROVIDED TO THE CITY PRIOR TO THE ISSUANCE OF A SUBSTANTIAL COMPLETION CERTIFICATE.
4. DELETERIOUS MATERIALS SHALL BE PREVENTED FROM ENTERING THE SEWER SYSTEM.
5. TOP OF INSPECTION CHAMBER STANDPIPES SHALL BE SET TO 600mm ABOVE FINAL LOT GRADE.
6. MANUFACTURED WYES OR APPROVED ALTERNATIVE SHALL BE USED FOR ALL SERVICE CONNECTIONS TO STORM MAINS.
7. PROVIDE A MINIMUM 1.5m COVER FOR STORM SEWERS.
8. ALL NEW DETENTION STRUCTURES, INFILTRATION FACILITIES AND FLOW CONTROL STRUCTURES EG. ORIFICES, GATES, WEIRS TO BE VIDEO TAPED OR PHOTOGRAPHED BY THE CONTRACTOR.
9. ALL PIPE TO BE PVC DR35 OR AS NOTED.

WATER

1. PRESSURE TESTING AND CHLORINATION OF NEW WATERWORKS TO BE PERFORMED PRIOR TO TIE-IN TO EXISTING WORKS AND WITNESSED BY ENGINEER OF RECORD. CONTRACTOR TO PROVIDE TESTING PLAN TO ENGINEER OF RECORD PRIOR TO UNDERTAKING TESTS.
2. THE CONTRACTOR SHALL ENSURE THAT ALL NEW WATERWORKS HAVE TEST POINTS AND TEMPORARY BLOW-OFFS SUITABLE TO ENSURE ADEQUATE PRESSURE TESTING, CHLORINATION AND FLUSHING. DISCHARGE OF CHLORINATED WATER INTO DITCHES, STORM SEWERS OR WATERCOURSES IS STRICTLY PROHIBITED.
3. PROVIDE A MINIMUM 1.8m COVER FOR WATERMAINS.
4. WATERMAINS SHALL BE MARKED BELOW GRADE USING A METALLIC DETECTABLE REINFORCED UNDERGROUND UTILITY MARKING TAPE. THE TAPE SHALL BE MINIMUM 150mm WIDE, METALLIC BLUE IN COLOUR AND SHALL BE MARKED "CAUTION: WATER LINE BURIED BELOW." THE TAPE IS TO BE INSTALLED ON TOP OF THE PIPE CUSHION, A MINIMUM OF 300mm ABOVE THE TOP OF THE PIPE. MARKING TAPE SHALL BE "THORTEC" OR APPROVED EQUAL.
5. A MINIMUM OF 1.5m HORIZONTAL CLEAR SEPARATION AND 150mm CLEAR VERTICAL SEPARATION SHALL BE MAINTAINED BETWEEN WATERMAINS AND ELECTRICAL CONDUITS, GAS MAINS AND TELEPHONE CONDUITS, UNLESS OTHERWISE SPECIFIED.
6. CONTRACTOR SHALL CONDUCT PRESSURE TEST IN ACCORDANCE WITH MMCD SPECIFICATIONS AND IN THE PRESENCE OF THE CIVIL ENGINEER.
7. CONTRACTOR TO MAINTAIN ACCESS AND WATER SERVICE TO LOCAL HOUSEHOLDS AND BUSINESSES AT ALL TIMES.
8. NEW WATERMAINS ARE TO BE PRESSURE TESTED, CHLORINATED AND BACTERIOLOGICALLY ACCEPTED BY INTERIOR HEALTH AUTHORITY PRIOR TO CONNECTION TO THE WATER SYSTEM. SUPERCHLORINATED WATER SHALL BE NEUTRALIZED IN ACCORDANCE WITH MINISTRY OF ENVIRONMENT AND DEPARTMENT OF FISHERIES AND OCEANS GUIDELINES PRIOR TO DISCHARGE TO ANY DRAINAGE COURSE.
9. HEALTH AUTHORITY PERMIT TO CONSTRUCT IS REQUIRED PRIOR TO COMMENCING WORK.
10. ALL PIPE TO BE PVC DR18 OR AS NOTED.

CROSSINGS⁵:

WATERMAIN ABOVE SEWER		>450mm CLEAR VERTICAL SEPARATION – NO SPECIAL PROTECTION REQUIRED
		1. 150 TO 450mm CLEAR VERTICAL SEPARATION – WATERMAIN JOINTS TO BE PROTECTED ⁷ – ADDITIONAL BEDDING STRUCTURAL SUPPORT REQUIRED
WATERMAIN BELOW SEWER ^{6,9}		2. >450mm CLEAR VERTICAL SEPARATION – BOTH WATERMAIN AND SEWER JOINTS TO BE PROTECTED ⁷
		3. 150 TO 450mm CLEAR VERTICAL SEPARATION – BOTH WATERMAIN AND SEWER JOINTS TO BE PROTECTED ⁷ – ADDITIONAL BEDDING STRUCTURAL SUPPORT REQUIRED

- ⁵WATERMAIN TO BE LAID ABOVE THE SEWER IF POSSIBLE, WITH THE CENTRE OF THE PIPE LOCATED AT THE CROSSING TO MAXIMIZE THE SEPARATION DISTANCE BETWEEN JOINTS. VERTICAL SEPARATION LESS THAN 150mm NOT ALLOWED.

⁷DISTANCE FOR PROTECTIVE MEASURES IS MEASURED NORMAL (PERPENDICULAR) TO THE SEWER. SEE TABLE 1 BELOW FOR PRECAUTIONS TO REDUCE THE RISK OF SANITARY AND STORM SEWAGE ENTERING WATERMAINS THAT MAY BE CONSIDERED.

⁸PROVIDE A HYDRAULIC BARRIER IN TRENCH BETWEEN SEWER AND WATERMAIN. BARRIER MUST EXTEND AT LEAST 300mm (1 FOOT) BEYOND OUTER EDGE OF WATERMAIN ON BOTH SIDES.

⁹NOT APPLICABLE TO SEWER FORCEMAINS.

TABLE 1. EXAMPLE PROTECTIVE MEASURES TO BE CONSIDERED (MORE THAN ONE MEASURE MAY BE REQUIRED)

– PROTECT WATERMAIN JOINTS AND SEWER JOINTS WITH SHRINK-WRAP, PETROLATUM TAPE, OR OTHER EQUIVALENT PIPELINE PROTECTION PRODUCTS. FOR EXISTING SEWERS, THE DISTANCE FOR PROTECTIVE MEASURES MAY BE REDUCED TO 1.5m (5 FEET) FROM NEW WATERMAIN TO AVOID EXCESSIVE EXCAVATION (SEE FIGURE 1 BELOW).
– CONSTRUCT SEWER OF EQUIVALENT CLASS PRESSURE PIPE OR REINFORCED CONCRETE PIPE USING FLEXIBLE GASKETS.
– ENCASE EITHER WATERMAIN OR SEWER INSIDE CASING PIPE SLEEVE. THE CASING PIPE MUST BE A MATERIAL THAT IS APPROVED FOR USE AS A WATERMAIN

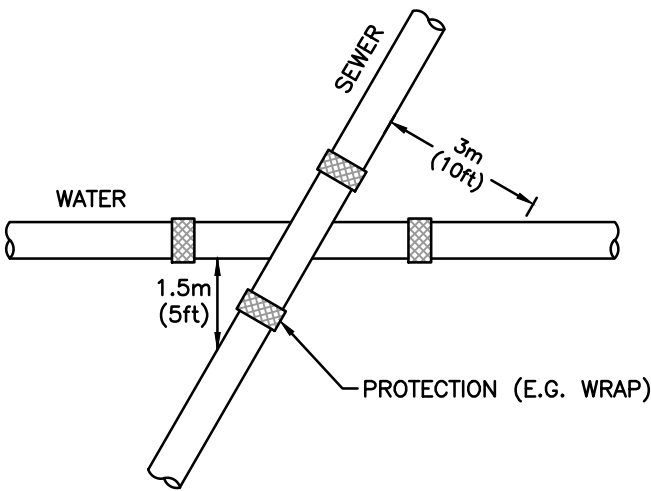
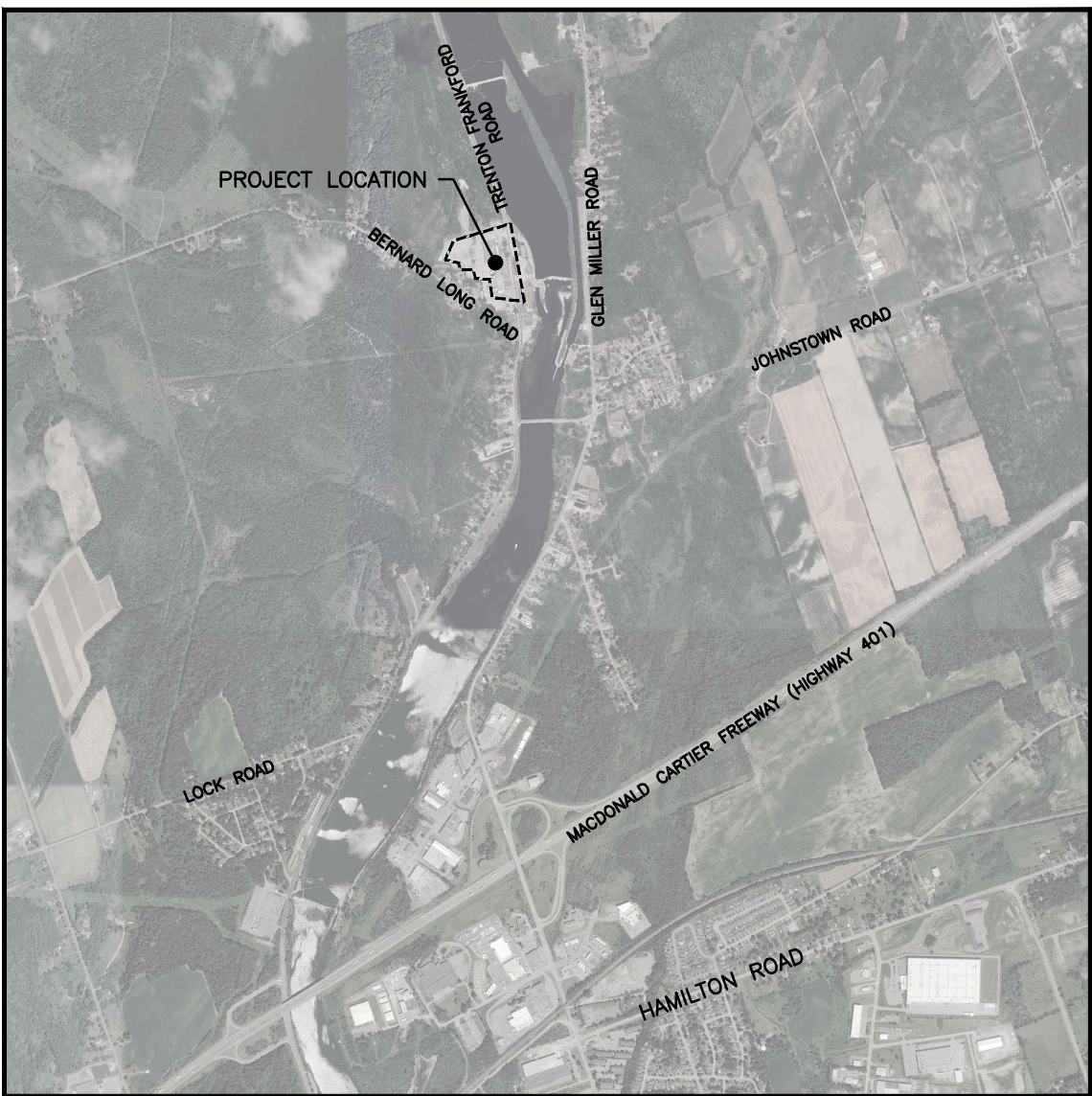
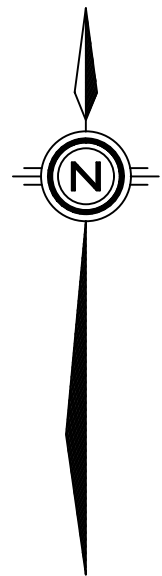
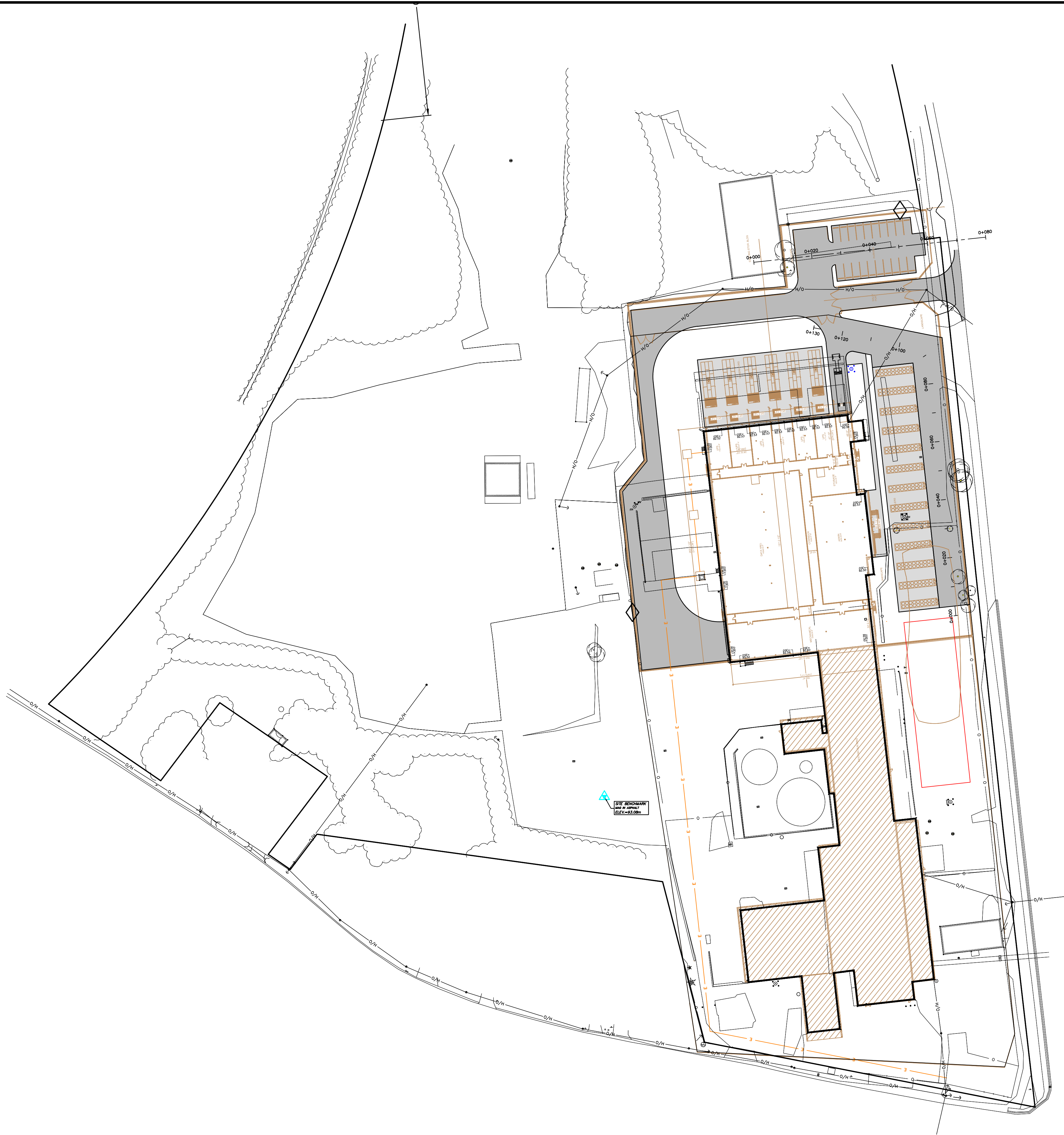


FIGURE 1. JOINT WRAPPING WHERE REQUIRED.

FRANCHISE UTILITIES

1. FRANCHISE UTILITIES INCLUDING BUT NOT LIMITED TO BC HYDRO, TELUS, SHAW, AND FORTIS ARE SHOWN FOR INFORMATION ONLY. REFER TO FRANCHISE UTILITY DRAWINGS SPECIFICATIONS FOR DESIGN AND CONSTRUCTION PURPOSES

DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.	SCALE AS NOTED DATE – APPROVED DSR DESIGN BY DSR DRAWN BY DB CHECKED BY DSR			PROJECT MANAGEMENT CIVIL ENGINEERING – FRANKFORD DATA CENTRE, QUINTE WEST, ON GENERAL NOTES	REV. No. A
2026-09-04	A	FOR APPROVAL – NOT FOR CONSTRUCTION	DB	DSR										PROJECT No. 2026-019
														DRAWING No. C001



KEY PLAN
NTS

DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	FOR APPROVAL - NOT FOR CONSTRUCTION	DB	DSR					



THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION.
THESE DRAWINGS SHOULD NOT BE SCALED. IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

SCALE AS NOTED
DATE -
APPROVED DSR
DESIGN BY DSR
DRAWN BY DB
CHECKED BY DSR

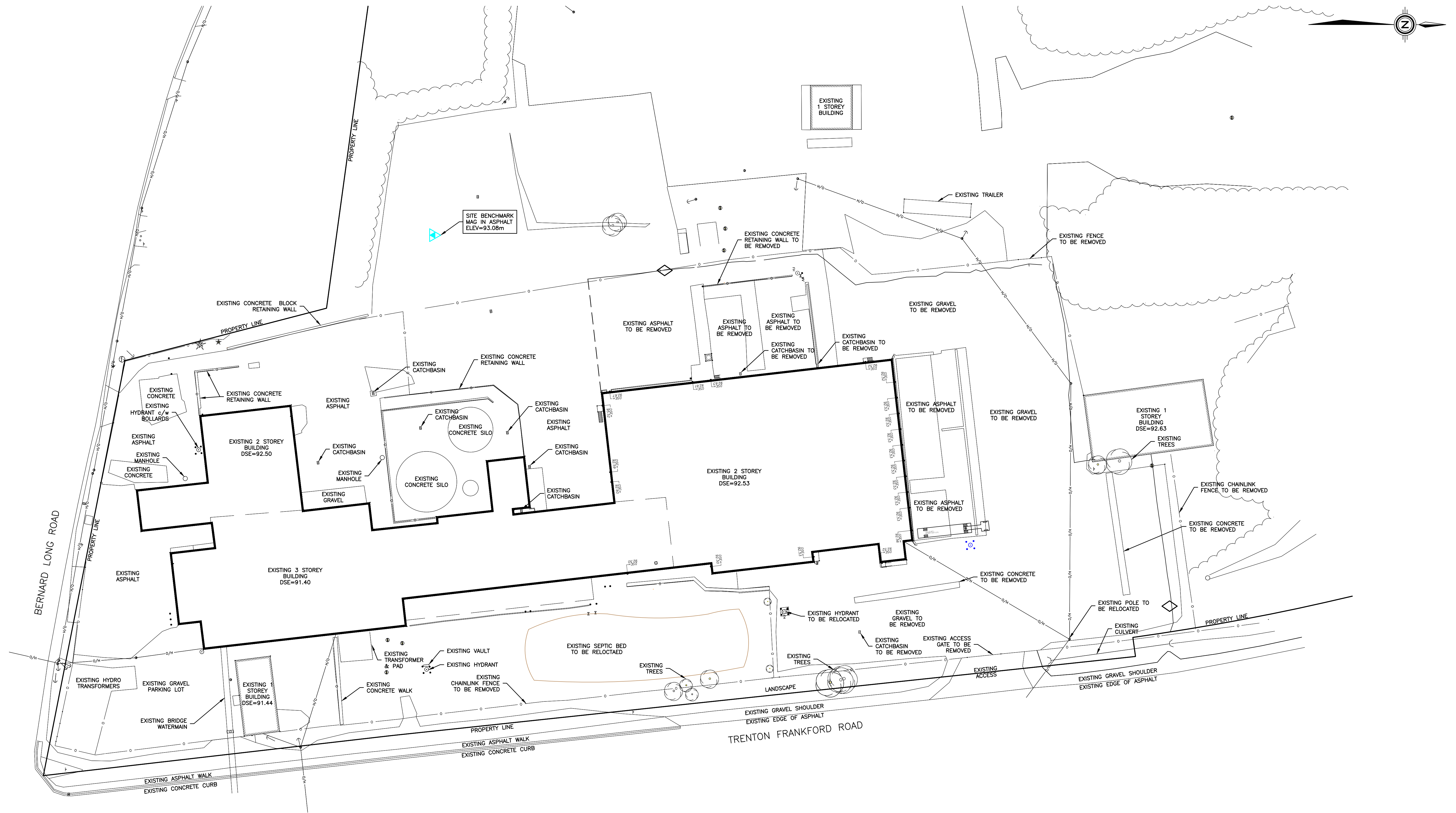


FUSION

PROJECT MANAGEMENT | CIVIL ENGINEERING

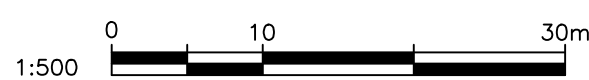
FRANKFORD DATA CENTRE, QUINTE WEST, ON
KEY PLAN

REV. No.	A
PROJECT No.	2026-019
DRAWING No.	C002

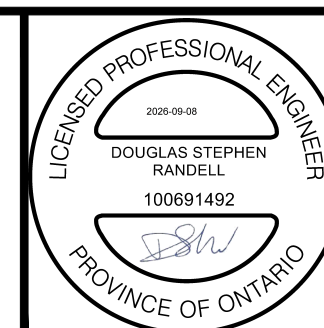


DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	FOR APPROVAL - NOT FOR CONSTRUCTION	DB	DSR					

THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION.
THESE DRAWINGS SHOULD NOT BE SCALED. IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

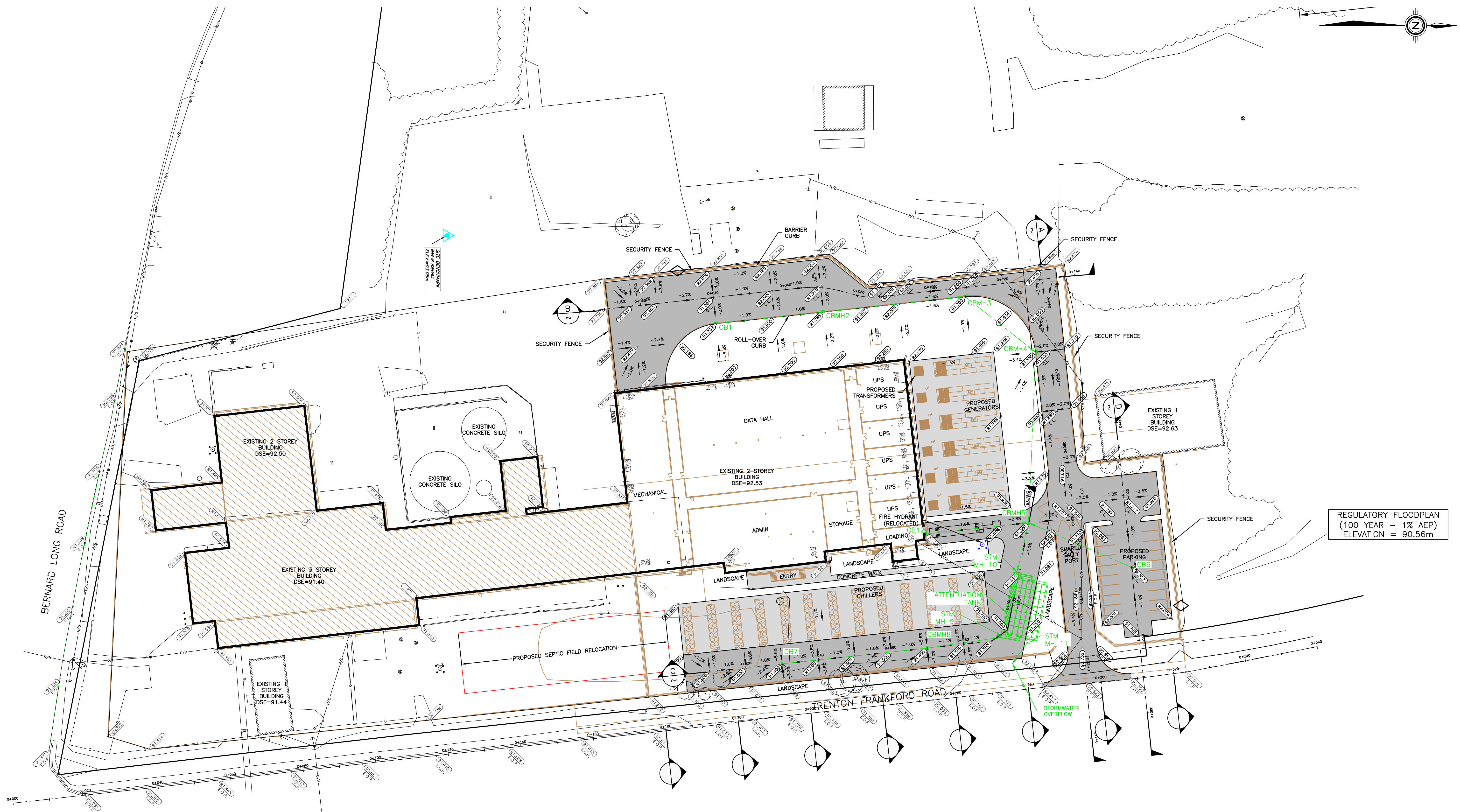


SCALE AS NOTED
DATE -
APPROVED DSR
DESIGN BY DSR
DRAWN BY DB
CHECKED BY DSR



FUSION PROJECT MANAGEMENT | CIVIL ENGINEERING
FRANKFORD DATA CENTRE, QUINTE WEST, ON
DEMOLITION PLAN

REV. No. **A**
PROJECT No. **2026-019**
DRAWING No. **C003**



DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	FOR APPROVAL - NOT FOR CONSTRUCTION	DB	DSR					



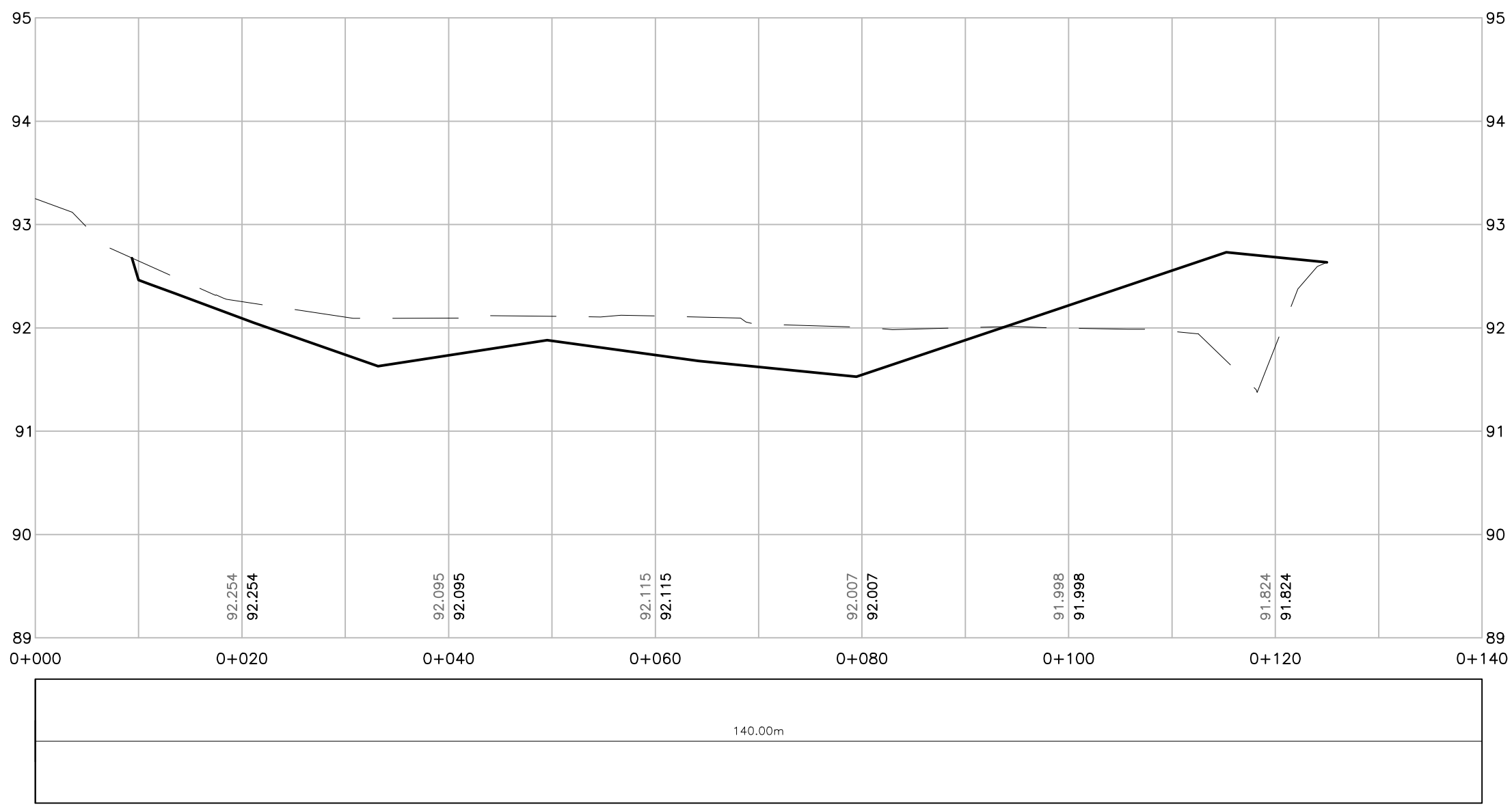
THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION.
THESE DRAWINGS SHOULD NOT BE SCALED. IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

SCALE AS NOTED
DATE -
APPROVED DSR
DESIGN BY DSR
DRAWN BY DB
CHECKED BY DSR

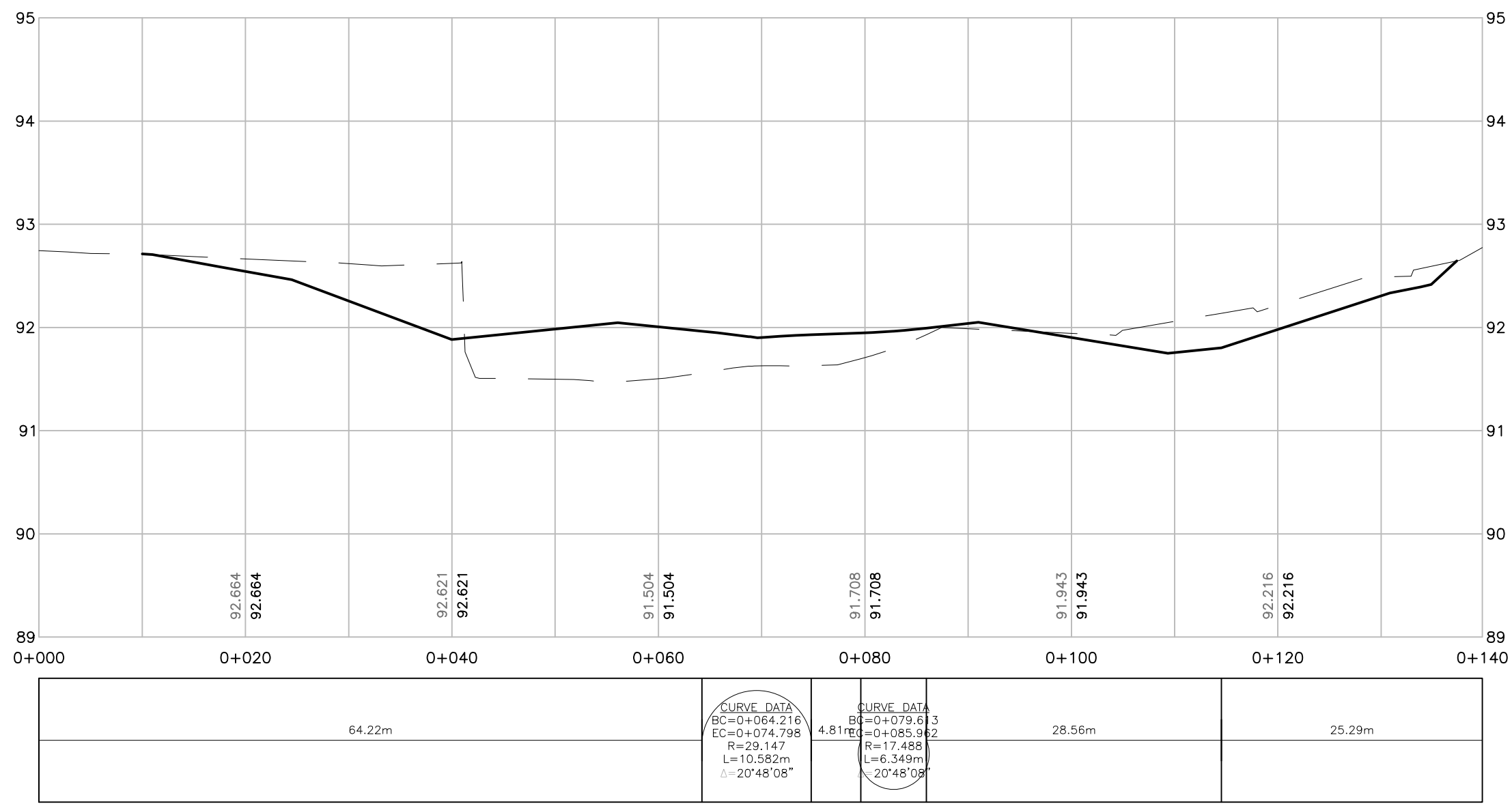


FUSION PROJECT MANAGEMENT | CIVIL ENGINEERING
FRANKFORD DATA CENTRE, QUINTE WEST, ON
GRADING PLAN

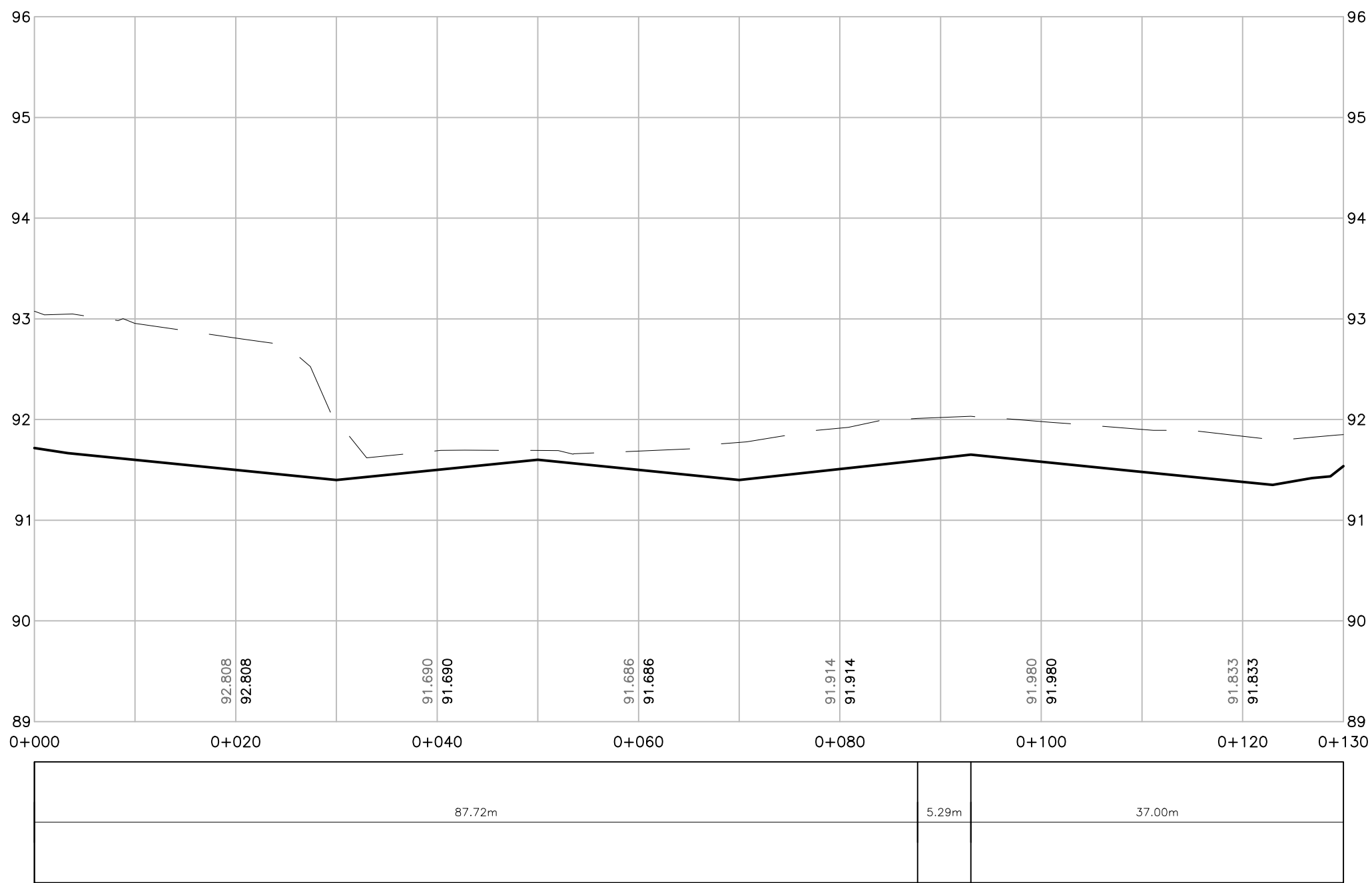
REV. No.	A
PROJECT No.	2026-019
DRAWING No.	C005



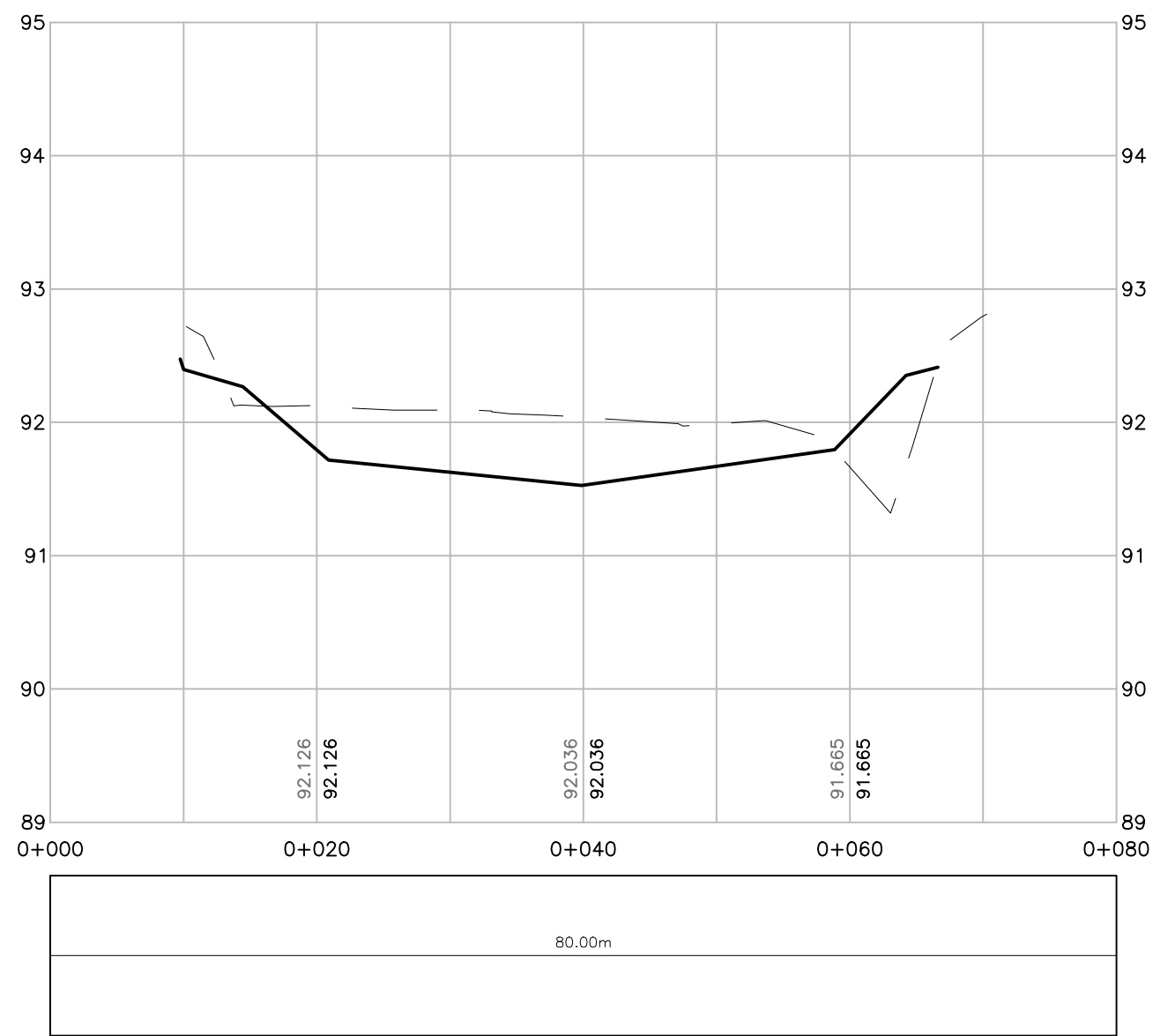
SECTION A



SECTION B



SECTION C



SECTION D

DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	FOR APPROVAL - NOT FOR CONSTRUCTION	DB	DSR					



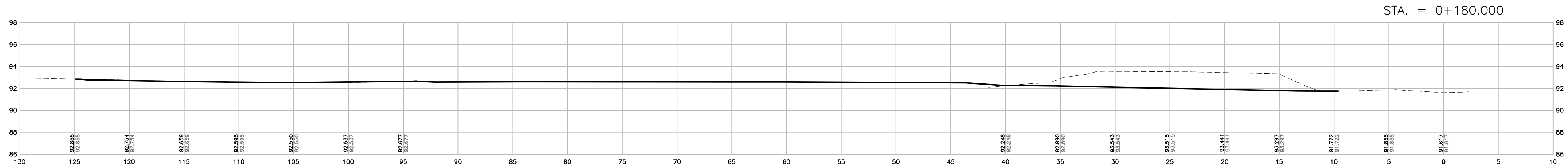
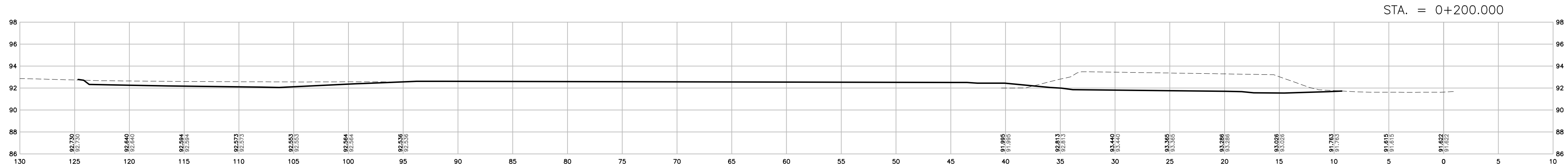
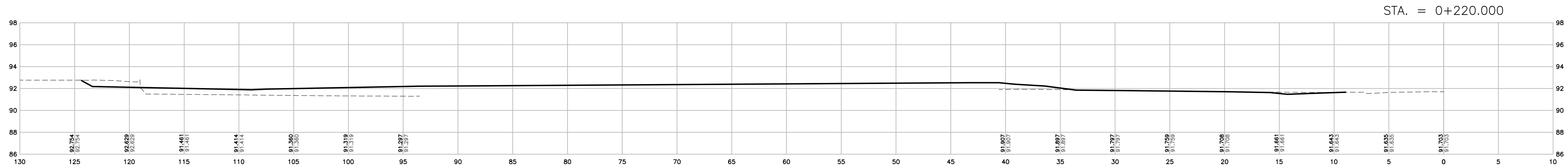
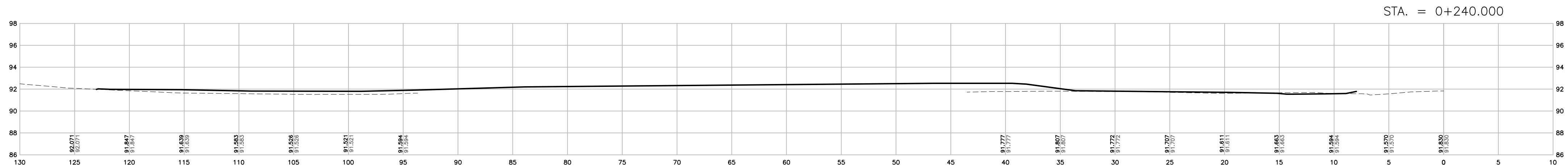
THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION. THESE DRAWINGS SHOULD NOT BE SCALED. IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

SCALE AS NOTED
DATE -
APPROVED DSR
DESIGN BY DSR
DRAWN BY DB
CHECKED BY DSR



FUSION PROJECT MANAGEMENT | CIVIL ENGINEERING
FRANKFORD DATA CENTRE, QUINTE WEST, ON
GRADING SECTIONS

REV. No.	A
PROJECT No.	2026-019
DRAWING No.	C006



DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	FOR APPROVAL - NOT FOR CONSTRUCTION	DB	DSR					

05m

1:250

THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION.
THESE DRAWINGS SHOULD NOT BE SCALED; IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

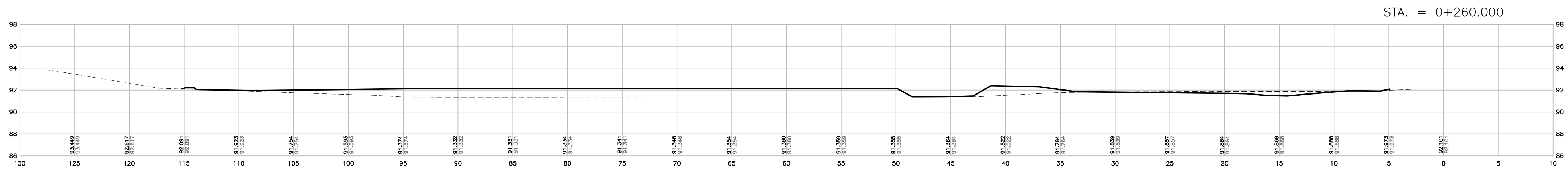
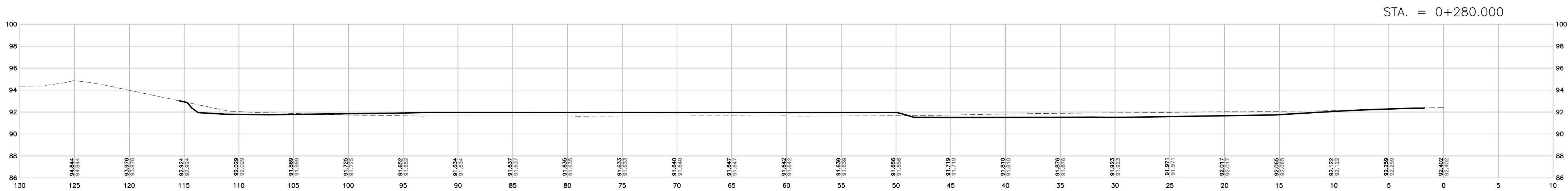
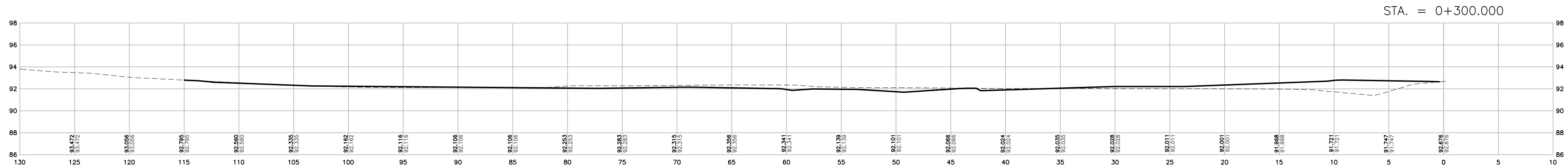
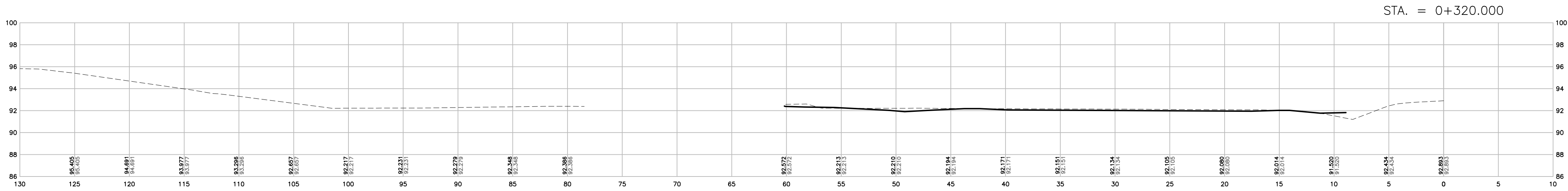
SCALE AS NOTED
DATE -
APPROVED DSR
DESIGN BY DSR
DRAWN BY DB
CHECKED BY DSR



PROJECT MANAGEMENT | CIVIL ENGINEERING

-
FRANKFORD DATA CENTRE, QUINTE WEST, ON
TRENTON FRANKFORD ROAD SECTIONS

REV. No.	A
PROJECT No.	2026-019
DRAWING No.	C007



DATE	REV	DESCRIPTION	BY	APPR.	DATE	REV	DESCRIPTION	BY	APPR.
2026-09-04	A	FOR APPROVAL - NOT FOR CONSTRUCTION	DB	DSR					

05m

1:250

THIS DRAWING AND DESIGN IS PROTECTED UNDER COPYRIGHT, AND ANY USE, REPRODUCTION OR REVISION IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION. THESE DRAWINGS SHOULD NOT BE SCALED. IF BEING USED FOR CONSTRUCTION, THE CONTRACTOR MUST CHECK AND VERIFY ALL DIMENSIONS, AND SHALL REPORT ANY DISCREPANCIES OR OMISSIONS PRIOR TO COMMENCEMENT OF ANY WORK.

SCALE AS NOTED

DATE -

APPROVED DSR

DESIGN BY DSR

DRAWN BY DB

CHECKED BY DSR

PROJECT MANAGEMENT | CIVIL ENGINEERING

-

FRANKFORD DATA CENTRE, QUINTE WEST, ON

TRENTON FRANKFORD ROAD SECTIONS

REV. No.	A
PROJECT No.	2026-019
DRAWING No.	C008

CITY OF QUINTE WEST

Office of the Mayor
Jim Harrison



7 Creswell Drive
P.O. Box 490
Trenton, ON K8V 5R6
TEL: (613) 392-2841
www.quintewest.ca

December 4, 2025

FirstBlock Inc.
2680A Bd Wilfrid-Hamel
Québec, QC
G1P 2J1

RE: Letter of Support for FirstBlock for a Sustainable Data Centre and Energy Recovery Project

To Whom It May Concern:

On behalf of the City of Quinte West, I would like to extend our support for FirstBlock Inc., an innovative Québec company that is redefining how data centres are integrated into the energy transition. Thanks to its high-efficiency liquid cooling and heat recovery technologies, FirstBlock transforms residual server heat into a renewable energy source that can be reused locally for heating buildings, industrial processes, or urban thermal networks. These projects contribute directly to the decarbonization objectives of Ontario and Canada, while generating regional economic benefits and strengthening the country's digital sovereignty through high-performance computing (HPC) infrastructure located on its territory.

Project description

Project name: Waste Heat Recovery Project on the 920 Trenton Frankford Site

Location: 920 Trenton Frankford Rd, Trenton, ON K8V 5P6

Description: The project consists in the construction of a high-temperature liquid-cooled data centre (55–65 °C outlet), capable of recovering and reusing more than 90% of the heat generated by the servers. The recovered heat will be used to support an agri-food hub: a vertical farm, greenhouses, and a fish and shrimp aquaculture facility, thereby reducing natural gas consumption and the associated GHG emissions.

From a municipal standpoint, this project will require all necessary permits prior to construction.

I wish to express my support for FirstBlock Inc. and for its efforts to develop eco-efficient data centres that serve the community, the climate, and the economic prosperity of Ontario and Canada. I invite government authorities to support its realization.

Sincerely,

Mayor Jim Harrison
City of Quinte West



HOUSE OF COMMONS
CHAMBRE DES COMMUNES

3700, rue du Campanile, bureau 110
Québec QC G1X 4G6
Tel: 418 648-3244

Joël Lightbound, Député de Louis-Hébert

Québec, January 28th 2025

Subject : Letter of Support for First Block Inc.'s Technological and Environmental Innovations

To Whom It May Concern,

I am writing to express my strong support for First Block Inc., a pioneering Canadian company that is redefining the future of sustainable data center infrastructures. Through cutting-edge innovations in high-performance computing (HPC) and heat recovery systems, First Block is addressing some of the most pressing challenges facing the data center industry while driving Canada's leadership in digital sovereignty, decarbonization, and technological innovation.

First Block Inc. exemplifies Canadian innovation at its best—combining cutting-edge technology, environmental stewardship, and economic impact. The company's solutions not only tackle the critical challenges of heat recovery but they also position Canada as a global leader in sustainable HPC technology. These innovations have the potential to transform the way data centers operate and how waste heat is utilized, creating lasting benefits for Canada's economy, environment, and digital infrastructure.

For the reasons outlined below by FirstBlock, I strongly endorse their initiatives and their efforts to advance sustainable development, digital sovereignty, and climate leadership.

Best regards,

Joël Lightbound, Louis-Hébert M.P.



HOUSE OF COMMONS
CHAMBRE DES COMMUNES

Technological Innovations

First Block has developed groundbreaking solutions that address key challenges in heat reuse and computing infrastructure, resulting in three distinct intellectual properties (IP) and patents for Canada :

1. **Cooling Plate Technology:** Advanced heat transfer systems designed to maximize thermal efficiency in computing chips.
2. **Dual Loop Design:** A rack system allowing for precise control of heat transfer while maintaining server performance.
3. **Solaris AI:** An intelligent optimization system designed to manage load distribution, analyze energy usage, and efficiently redistribute resources in liquid-cooled data centers.

These innovations have been validated through the **RSDE (Scientific Research and Experimental Development) report**, further cementing First Block's position as a leader in sustainable HPC technology.

Revolutionizing Heat Management in Data Centers: Turning Waste into Opportunity

Data centers, the backbone of the digital economy, are evolving rapidly to meet the surging demand for computational power. However, this evolution comes with significant challenges in heat management due to:

- **Increased Density:** As computing hardware becomes more powerful, the density of heat generated per square foot rises dramatically, straining traditional cooling systems.
- **Outdated Cooling Technologies:** Many data centers still rely on conventional cooling methods that are inefficient and unsuited for the needs of modern high-performance computing (HPC).
- **Heat Treated as Waste:** In most facilities, the vast amount of heat generated by servers is discarded, representing a missed opportunity to contribute to sustainability goals and resource efficiency.

The Challenges of Heat Reuse

To transform data centers into sustainable energy hubs, two critical barriers must be addressed:

Challenge 1: Managing Low-Temperature Heat (60°C)

Traditional cooling systems often result in exit temperatures too low for practical reuse in industrial, residential, or district heating applications.



HOUSE OF COMMONS
CHAMBRE DES COMMUNES

Challenge 2: Aligning Heat and Computational Demands

The unpredictable and variable demands of computing loads make it difficult to maintain an efficient and consistent heat output for downstream applications.

First Block's Innovations

By tackling these challenges head-on, First Block has developed groundbreaking solutions that redefine the role of data centers in the energy ecosystem:

- **Higher Output Temperatures:** Using advanced plate heat exchangers and dual loop cabinet technology, First Block has increased heat output to 80°C, making it suitable for a wide range of applications.
- **Dynamic Heat Adaptation:** With Solaris AI, heat output adjusts in real time to match the variable demands of computing loads and energy users, ensuring efficiency and waste minimization.

From Waste to Resource

These innovations transform heat from a wasted byproduct into a valuable resource, positioning data centers as pivotal players in sustainable energy systems. With this approach, First Block is enabling Canada to lead in the global push for greener, more efficient data center operations, driving both economic and environmental benefits.

Strategic Benefits for Canada

1. Innovation Leadership

First Block's technologies represent a significant step forward in sustainable data center operations. By integrating patented solutions, the company is setting new global standards for heat recovery and energy efficiency.

2. Digital Sovereignty

With HPC becoming a geopolitical priority, First Block's projects directly support Canada's sovereignty by reducing reliance on foreign-controlled data centers and establishing local infrastructure for critical computing needs.



HOUSE OF COMMONS
CHAMBRE DES COMMUNES

3. Decarbonization

The reuse of waste heat from HPC data centers aligns with Canada's climate goals, eliminating fossil fuel dependency and cutting CO₂ emissions across industrial, residential, and district heating sectors.

4. Economic and Environmental Value

First Block's heat recovery technology transforms waste into a valuable resource, providing sustainable heating solutions for industries, municipalities, and homes while reducing energy costs.

CITY OF QUINTE WEST – COUNCIL MEETING OF OCTOBER 7, 2026

Proposed Exemption – By-law 26-131, Interim Control By-law, Data Centres (Item 16.1, Mayor Proposed By-laws)

1. Purpose

By-law 26-131, as placed on the October 7, 2026 agenda, prohibits any Data Centre use within the Interim Control Area for one year. Section 4 excepts only a Data Centre use that lawfully existed on the date of passage. It contains no exemption for lands in a zone that permits a Data Centre and for which an application has already been received by the City and deemed complete under the *Planning Act*.

The proposed addition below would allow the City to complete the study contemplated by the By-law while applications already in the City's planning process, on lands already zoned to permit the use, continue to be reviewed on their own merits, subject to conditions. It is offered for the Mayor's consideration, to be incorporated in the By-law as tabled or in a revised By-law placed before Council.

2. Section 4 of By-law 26-131 as tabled

4. Notwithstanding any other By-law to the contrary, no person shall, within the Interim Control Area, use any land, building or structure including any expansion or addition to an existing use, building or structure, for the purpose of a Data Centre, except for a Data Centre use that lawfully existed on the date of the passage of this By-law, as long as it continues to be used for such purpose.

3. Proposed addition

THAT By-law 26-131 be revised by adding the following sections after Section 4:

4A. Notwithstanding Section 4, this By-law shall not apply to lands that satisfy both of the following conditions on the date of passage of this By-law:

- (a) the lands are in a zone under the City of Quinte West Comprehensive Zoning By-law 14-86, as amended, that permits a Data Centre; and
- (b) an application under the *Planning Act* relating to a Data Centre on the lands has been received by the City and deemed complete,

and such applications shall continue to be processed in accordance with the City's established planning process, each on its own merits.

4B. In evaluating any application referred to in Section 4A, the City shall require that the Data Centre:

- (i) not consume municipal water for cooling;
- (ii) comply at all times with the Ministry of the Environment, Conservation and Parks noise guidelines (NPC-300), supported by an acoustic assessment prepared by a qualified professional engineer;
- (iii) make waste heat available for recovery by neighbouring industrial or agricultural users through a heat-exchanger connection;

- (iv) operate any emergency backup generation in compliance with the *Environmental Protection Act*, the *Clean Water Act, 2006* and all other applicable federal and provincial requirements for fuel storage and emissions;
- (v) be supplied from the existing provincial electricity grid, with any connection or system upgrade costs attributable to the facility borne by the proponent in accordance with Ontario Energy Board cost-responsibility rules; and
- (vi) demonstrate alignment with Ontario's Data Centre Playbook and Canada's Responsible Data Centre Development Principles, including their objectives of data security and Canadian digital sovereignty.

4C. Nothing in Sections 4A or 4B constitutes approval of any application, and each application remains subject to all applicable requirements of the *Planning Act*, the City's Official Plan and Zoning By-law, and the ordinary approval authority of Council or its delegate.